

Quantifying Heterogeneity in Bovine Tuberculosis Breakdown Dynamics Using Clustering and Finite Mixture Modelling

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Abstract

Bovine tuberculosis (bTB) remains one of the most costly and persistent animal health challenges in Ireland. This study applied unsupervised machine learning to identify distinct epidemiological phenotypes of bTB breakdowns using herd-level surveillance data.

Five variables capturing breakdown duration, index test severity, within-herd spread, breakdown history, and herd size were log-transformed and standardised prior to analysis. Partitioning Around Medoids (PAM) was the primary clustering method, with k-means applied as a comparator. Three epidemiologically distinct phenotypes were identified. Cluster 1 represented the modal breakdown, characterised by moderate severity and limited within-herd prevalence. Cluster 2 represented a "quiet herd" phenotype with small, low-severity breakdowns yet the highest rate of prior breakdown history, consistent with residual infection or continual low-level environmental exposure. Cluster 3 represented an explosive phenotype associated with longer durations, elevated within-herd prevalence likely explained by high within-herd spread, and greater transmission burden, consistent with naïve herds encountering intense infection pressure.

Multinomial regression demonstrated that neighbourhood infection pressure particularly the presence of new breakdown herds locally was the dominant predictor of cluster membership across all three phenotypes relative to control herds, suggesting a microepidemic phenomenon in which the surrounding epidemiological context is a stronger determinant of breakdown phenotype than herd-level characteristics alone. Survival analysis confirmed significant differences in time to subsequent breakdown (log-rank $p < 0.0001$), with Cluster 2

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showing the fastest recurrence and Cluster 1 the longest breakdown-free periods; Cox proportional hazards modelling confirmed these differences persisted after covariate adjustment.

These findings demonstrate that bTB breakdowns are epidemiologically heterogeneous, reflecting distinct dynamics of infection, persistence, and reinfection. Phenotype-informed intervention intensified environmental investigation for recurrent low-level herds and rapid response for explosive phenotypes represents a more targeted and resource-efficient approach to bTB control.

Keywords: Bovine tuberculosis, Mycobacterium bovis, Clustering, PAM, Epidemiology, bTB Breakdown Heterogeneity

1. Introduction

Bovine tuberculosis (bTB), caused by *Mycobacterium bovis*, represents a persistent challenge for Ireland’s agricultural sector. Eradication matters greatly for Ireland’s standing in international trade, as countries with official tuberculosis-free status gain competitive advantages in selling livestock and dairy products globally [1]. The disease additionally carries significant zoonotic risks: humans can contract tuberculosis from unpasteurised dairy products and through direct contact with infected cattle, making bTB a genuine occupational hazard for farmers and veterinary workers [2].

Ireland operates within a complex European regulatory framework that shapes its bTB control strategy. Cattle vaccination remains illegal across the EU, restricting Ireland’s policy options and preventing adoption of vaccination as a supplementary control measure despite experimental evidence of potential benefit [3]. The EU’s broader framework for tuberculosis in wildlife recognises that bTB persistence in countries such as Ireland reflects the presence of wildlife reservoirs and demands integrated surveillance across domestic and wild animal populations [4]. As a non-tuberculosis-free member state, Ireland must maintain rigorous testing and surveillance protocols mandated by EU regulation, generating ongoing financial and administrative burdens that will persist until eradication is achieved and official free status is granted [5].

The financial consequences of bTB extend well beyond individual farm losses, affecting entire agricultural systems and national economies. Control of bTB in the Republic of Ireland costs approximately €84 million annually [6], though this figure likely underestimates the true economic burden when indirect farm-level losses are accounted for. For individual farmers, the primary economic impact arises from testing costs and movement restrictions imposed when herds are under investigation or confirmed as infected. Ireland bears these substantial costs year after year without achieving eradication [7].

Ireland’s persistent bTB challenge reflects three interconnected epidemiological barriers that collectively confound eradication efforts. Wildlife reservoirs, particularly the European badger (*Meles meles*), maintain infection within the landscape and serve as a continuous source of transmission to cattle populations, undermining the impact of cattle-focused control measures alone [8]. Residual infection, where infected cattle remain undetected due to imperfect diagnostic sensitivity, creates silent reservoirs within herds that perpetuate transmission cycles and seed recurrent breakdowns [9]. Cattle trade movements introduce infected animals often undetected carriers across regions, with genomic evidence demonstrating that inward cattle movements are a primary driver of infection in high-risk herds [10]. Addressing bTB eradication therefore demands simultaneous intervention across all three routes: enhanced detection of infected cattle, management of wildlife reservoirs, and risk-based restrictions on cattle trade to prevent geographic spread [7]. Without a comprehensive strategy targeting wildlife, residual infection, and movement simultaneously, Ireland’s national eradication programme will remain constrained [10].

Bovine tuberculosis breakdowns in Irish herds do not follow a single epidemiological pattern but instead vary considerably in their characteristics, with surveillance records demonstrating substantial differences in breakdown frequency, severity, duration, and likelihood of recurrence across herds and over time [11, 12]. Some herds experience small, self-limiting outbreaks involving few reactor animals that resolve rapidly following detection, while others face serious within-herd spread, prolonged movement restriction periods, or repeated breakdown events in close succession [13]. These differences are not random fluctuations around a common mean but suggest that distinct underlying processes drive different breakdown types, such that a single disease process model is insufficient to characterise the full range of observed outbreak behaviour.

This variation likely arises from the interplay of multiple factors operating at different epidemiological levels. Within herds, differences in infection burden, immune status, and the sensitivity of ante-mortem diagnostic tests determine the apparent size of an outbreak and the duration over which infection persists before clearance can be achieved [11]. Between herds, inward cattle movements carrying undetected infection introduce the pathogen to new farms and across regional boundaries, while wildlife interactions and environmental contamination determine whether infection becomes established and how long it persists following initial detection [14]. Different combinations of these within-herd and between-herd factors produce breakdown profiles with distinct risk characteristics and fundamentally different requirements for effective control.

From a disease management perspective, recognising this heterogeneity is essential for the efficient use of finite control resources. Small, contained outbreaks require fundamentally

different intervention responses than large, persistent ones or herds experiencing repeated recurrence in close succession [15]. A programme that applies uniform responses regardless of breakdown type risks both under-resourcing the most complex cases and over-investing in those that would resolve rapidly under standard protocols. A clearer characterisation of breakdown phenotypes would allow the national eradication programme to direct intensified surveillance, enhanced biosecurity support, and targeted wildlife management toward the herds and breakdown types where intervention is most likely to have impact [15].

Classifying breakdowns into groups with similar epidemiological profiles moves beyond population-level averages to reveal the genuine structure present in national surveillance data. Data-driven classification approaches can identify meaningful and reproducible categories based on observed breakdown characteristics without imposing prior assumptions about what patterns should exist or how many distinct phenotypes are present [11]. This opens the door to examining temporal trends in phenotype prevalence, regional variation in breakdown character, and phenotype-specific associations with risk factors. Understanding this heterogeneity is a critical foundation for characterising how bTB spreads and persists within Irish cattle populations, and for designing control strategies that are commensurate with the true diversity of outbreak behaviour rather than optimised for an average herd that may represent few actual cases.

Clustering groups bTB breakdowns by their shared characteristics without forcing pre-determined categories onto the data [11]. This approach works well when breakdown types have not been clearly defined, allowing researchers to discover patterns naturally present in surveillance records [16]. This sorted classification then supports systematic study of how these breakdown types change over time and vary across different regions [15]. Clustering therefore gives researchers a clear, data-driven way to understand and explain the different forms that bTB takes across Irish herds [17].

Based on the need to better understand heterogeneity in bTB breakdowns, the objectives of this study were:

1. To quantify and characterise heterogeneity in bTB breakdowns using clustering techniques and finite mixture modelling applied to breakdown-level surveillance data.
2. To describe trends in these clusters overtime.
3. To identify risk factors for each cluster type.
4. To quantify survival to a subsequent breakdown (time to failure) depending on cluster type.

2. Materials and Methods

2.1. Data sources

Data used in this study were derived from Ireland’s national bovine tuberculosis (bTB) surveillance infrastructure, maintained as part of the country’s ongoing eradication programme [1]. Three interlinked administrative datasets were used. The national bTB herd breakdown database provided longitudinal records of all confirmed bTB breakdown events in Irish cattle herds, including breakdown initiation date, duration, test results at the index and subsequent rounds of testing, and reactor counts disaggregated by animal category. The Animal Identification and Movement System (AIMS) provided records of all notified cattle movements between registered herds, enabling characterisation of the movement history of each breakdown herd in the period preceding the index event [18]. A neighbourhood exposure dataset derived from spatial herd records provided counts of epidemiologically relevant neighbouring herds those under active restriction, those that had recently cleared restriction, and those with no recorded breakdown history within a defined proximity of each index herd [19]. All datasets were subject to data validation and completeness checks prior to linkage, and were linked at the level of the individual herd breakdown event using a unique breakdown identifier.

2.2. Cohort definition and breakdown-level filtering

To identify a cohort of epidemiologically interpretable breakdown events suitable for unsupervised cluster analysis, a series of pre-specified exclusion criteria were applied to the national breakdown database. Breakdowns were excluded if they recorded zero confirmed reactor animals, as these events are unlikely to represent genuine active infection. Breakdowns with a total duration of fewer than 60 days were excluded on the grounds that very short events may reflect administrative or procedural closures rather than biologically meaningful disease dynamics. Breakdowns occurring within herds designated as controlled finishing units (CFUs) were excluded, as CFUs operate under distinct biosecurity and movement conditions that confound comparison with conventional commercial herds. To ensure epidemiological comparability of the index event, only breakdowns initiated either by routine slaughter surveillance or single intradermal comparative tuberculin test (SICTT) using tuberculin skin test types 1 or 8 were retained.

2.3. Construction of clustering variables

Five variables were selected to capture complementary epidemiological characteristics of breakdown behaviour at the herd level. Variable selection was guided by biological plausibility, data completeness, and the requirement to avoid redundancy within the clustering

feature set. Continuous variables with right-skewed distributions were log-transformed using the $\log(1 + x)$ transformation prior to analysis to reduce the influence of extreme values. All variables were subsequently standardised to zero mean and unit variance to ensure that no single variable dominated the distance calculations by virtue of its scale.

- **Severity at detection** (`prop_positive_index`): the proportion of animals testing positive at the index test, calculated as the number of confirmed index reactors divided by the total number of animals tested at the index round. This variable captures the apparent prevalence of infection within the herd at the point of breakdown detection.
- **Within-herd transmission** (`spread_ratio`): the ratio of reactors identified at subsequent testing rounds to reactors identified at the index test. Values greater than one indicate progressive amplification of detected infection within the herd following the initial test, reflecting ongoing within-herd transmission or test sensitivity effects. Values of zero indicate no additional reactors were identified after the index round.
- **Breakdown persistence** (`bd_duration_days`): the total duration of the breakdown in days, measured from the date of breakdown initiation to the date of final clearance. This variable reflects the combined effect of infection burden, test sensitivity, and management response on the time required to achieve herd clearance.
- **Long-term breakdown history** (`no_bds_started_in_prev_5yr`): the number of distinct breakdown events initiated within the same herd in the five-year period preceding the index breakdown. This variable captures chronic or recurrent susceptibility, reflecting persistent environmental, wildlife, or management risk factors at the herd level.
- **Herd size** (`herd_size_mean`): the mean of herd size recorded at three seasonal time points within the breakdown year (January, May, and September), providing a temporally stable estimate of herd scale. Herd size influences the absolute number of animals at risk, the statistical power of the skin test to detect infection, and the logistical complexity of clearance.

Two candidate variables considered during variable selection were excluded from the final clustering feature set on analytical grounds. The binary within-herd spread indicator (`spread_any`) was excluded as it is a direct binary transformation of `spread_ratio` and its inclusion would double-weight the spread dimension in distance calculations. The short-term recurrence indicator (`bd_within_13_months`), which flags whether a new breakdown occurred within 13 months of the preceding clearance, was excluded because it is derived

from the inter-breakdown interval used as the outcome variable in the subsequent survival analysis; its inclusion in the clustering feature set would have introduced circularity into the evaluation of cluster differences in time to subsequent breakdown. Both variables were retained in the dataset and used as covariates in downstream analyses.

2.4. Handling of Missing Data

No global imputation procedure was applied. Variables representing prior breakdown history were assigned a value of zero where no preceding events were recorded, reflecting the structural absence of prior exposure rather than a missing observation. Derived variables were constructed using conditional logic to prevent undefined values: proportions and ratios were set to missing where the denominator was zero, and infinite values arising from division were recoded as missing prior to analysis. Records with missing values in any of the five clustering variables were excluded from the analytical cohort, ensuring that cluster structure was determined exclusively by observed data without recourse to imputed values.

2.5. Data Transformation, Standardisation, and Exploratory Analysis

Continuous clustering variables with right-skewed distributions were subjected to a $\log(1+x)$ transformation prior to analysis to reduce the influence of extreme values and stabilise variance. All five transformed variables were subsequently standardised to zero mean and unit variance, ensuring that no variable exerted disproportionate influence on distance calculations by virtue of its scale alone.

Exploratory data analysis was conducted on the full analytical cohort to characterise marginal distributions, assess inter-variable relationships, and identify potential redundancy prior to clustering. Raw and log-transformed distributions were examined using histograms and kernel density estimates. Pairwise relationships among the standardised clustering variables were assessed through a scatter plot matrix and Pearson correlation matrix. The joint distribution of index test severity and within-herd spread ratio was additionally visualised to characterise the relationship between these two epidemiological dimensions. Boxplots of the standardised variables were examined to identify outlying observations. All exploratory figures are provided in the Supplementary Material. Subsampling was applied only where computationally necessary: 2,000 observations for the Hopkins statistic and visual assessment of clustering tendency, and 5,000 observations for cluster number selection criteria and silhouette visualisations. All canonical clustering and downstream analyses were performed on the complete dataset.

2.6. Principal Component Analysis

Principal Component Analysis (PCA) was applied to the standardised matrix of five clustering variables prior to clustering to characterise the latent structure of the data, examine inter-variable relationships in a reduced-dimensional projection, and provide a visualisation space for assessing cluster separation. PCA was performed on the pre-scaled feature matrix without additional centring or scaling. The proportion of variance explained by each component was examined to assess effective dimensionality, and the contribution of individual variables to the first and second principal components was inspected to aid epidemiological interpretation of the dominant axes of variation. PCA scores were subsequently used to visualise cluster assignments in the plane defined by the first two principal components. All PCA computations were performed on the complete analytical cohort. Variable contribution plots, the biplot, and cluster visualisations in PCA space are provided in the Supplementary Material.

2.7. Cluster analysis

2.7.1. Assessment of clustering tendency

Prior to clustering, the tendency of the data to exhibit genuine cluster structure was assessed using two complementary approaches. The Hopkins statistic was computed on a random subsample of 2,000 observations, with values approaching 1 indicating strong departure from spatial uniformity and the presence of genuine clustering structure. The Visual Assessment of cluster Tendency (VAT) method was additionally applied, producing an ordered dissimilarity matrix in which block-diagonal structure provides visual evidence of cluster presence (Supplementary Material 1 Figure 12).

2.8. Clustering method

2.8.1. Partitioning Around Medoids (PAM)-Primary Method

The primary clustering method was Partitioning Around Medoids (PAM), applied to the full standardised feature matrix comprising all eligible breakdown events. PAM was selected in preference to k-means for two principal reasons. First, PAM identifies cluster representatives as actual observed breakdown events (medoids) rather than arithmetic centroids, making cluster prototypes directly interpretable as real epidemiological units whose characteristics can be examined and reported. Second, PAM minimises the sum of dissimilarities between observations and their assigned medoid rather than the sum of squared Euclidean distances, conferring greater robustness to outlying observations than k-means.

The optimal number of clusters was determined by convergent evidence across four independent criteria, each applied to a random subsample of 5,000 observations to reduce com-

putational cost. The within-cluster sum of squares (WSS) elbow method and the variance-explained elbow method, implemented via the ClusterR package, were used to identify the value of k beyond which additional clusters yield diminishing reductions in intra-cluster heterogeneity. The average silhouette width criterion identified the value of k that maximised the mean ratio of between-cluster separation to within-cluster cohesion. Majority vote across 26 validity indices implemented in the NbClust package provided a multi-criterion summary recommendation. The gap statistic was additionally computed; given its known tendency to favour $k = 1$ in large, overlapping datasets, it was treated as supplementary rather than primary evidence. The PAM algorithm was subsequently executed with $k = 3$ on the complete standardised dataset. The resulting cluster assignments constituted the canonical solution used throughout all downstream analyses. Visualisation of cluster separation for $k = 2$ to $k = 5$ in principal component space and corresponding silhouette profiles are provided in Supplementary Material 1 Figures 4 and 13, respectively.

2.8.2. K-means Clustering-Comparator Method

K-means clustering was applied as a comparator to evaluate the consistency and robustness of the PAM solution. In k-means, k cluster centroids are initialised and iteratively updated by assigning each observation to the nearest centroid and recalculating centroids as the mean of assigned observations, continuing until within-cluster sum of squares converges. To avoid local minima, 25 random restarts were used and the solution with lowest total WSS was retained. K-means was applied for $k = 2, 3, 4$, and 5 on the same scaled matrix used for PAM, and additionally within the reduced-dimensional space defined by the first three principal components, providing an exploratory assessment of cluster structure in a lower-dimensional projection of the feature matrix.

Pairwise agreement between k-means and PAM solutions at each value of k was quantified using the Adjusted Rand Index (ARI), where $ARI = 1$ denotes perfect agreement and $ARI = 0$ denotes agreement at chance level. Average silhouette widths were computed for both methods across $k = 2$ to $k = 5$ and compared directly. The final cluster solution was based on PAM throughout all downstream analyses.

2.9. Additional clustering methods

As well as PAM and K-means, alternative methods were also trialed, including Gaussian Mixture Model (GMM), Fuzzy C-means, and Consensus Clustering. Method and results for these approaches are included in Supplementary Material 2.

2.10. Multinomial Regression

To investigate the epidemiological determinants of breakdown cluster membership, a multinomial logistic regression model was fitted with cluster assignment as the polytomous outcome. The analysis sought to identify which movement and neighbourhood exposure variables were independently associated with an increased probability of belonging to each of the three breakdown phenotypes relative to herds with no recorded breakdown history.

Herds with no recorded breakdown during the study period were assigned to the reference category (Cluster 0), providing a clinically and epidemiologically meaningful baseline against which the three breakdown phenotypes identified through PAM clustering and characterised by distinct profiles of infection severity, within-herd spread, duration, and recurrence could be contrasted. The multinomial logistic regression framework extends binary logistic regression to accommodate a polytomous outcome by simultaneously estimating the log-odds of belonging to each breakdown cluster relative to the reference category, yielding a set of cluster-specific coefficient vectors that characterise the independent contribution of each predictor to the probability of breakdown cluster membership.

This analytical approach allows direct comparison of the relative importance of trade-related exposure through inward cattle movements from herds of differing bTB status and spatially mediated neighbourhood infection pressure, providing insight into the mechanisms through which herd-level epidemiological context shapes not only whether a breakdown occurs but what form that breakdown takes.

2.11. Data Preparation for multinomial regression

The analytical dataset was constructed by combining two sources. The first comprised herds with confirmed bTB breakdowns assigned to one of three PAM-derived clusters (Clusters 1–3). The second comprised control herds those with no breakdown history over the observation period assigned to Cluster 0. A random sample of 30,000 control observations was drawn to ensure computational tractability while preserving representativeness of the control population.

2.11.1. Predictor Variables

Seven predictor variables were included, capturing two complementary domains of epidemiological exposure to *M. bovis* infection.

The first domain comprised four movement-related variables, each reflecting the bTB status of the source herd from which cattle were received during the observation period: inward movements from Control herds (**Control_moves**), New Breakdown herds (**NewBD_moves**), Burnout herds (**Burnout_moves**), and herds of Unknown status (**Unknown_moves**). Herd status classifications were defined as follows. Control herds were those with no confirmed

bTB breakdown in the index year, the year immediately preceding the index year, or the year immediately following. New Breakdown herds were those with a confirmed breakdown initiated in the index year. Burnout herds were those that did not meet the criteria for either Control or New Breakdown status, typically representing herds in the later stages of an ongoing restriction or recently cleared from restriction. Unknown herds were those to which a definitive bTB status could not be assigned based on available records.

The second domain comprised four neighbourhood exposure variables, each quantifying the number of spatially proximate herds those within a defined spatial radius of the index herd classified according to the same four status categories in the current year: `Burnout_neighbours`, `Control_neighbours`, `NewBD_neighbours`, and `Unknown_neighbours`. Neighbourhood variables capture indirect spatial transmission risk arising from local infection pressure, independently of any direct animal trade linkage between herds.

Together, the two domains characterise both direct trade-mediated exposure through the introduction of animals from herds with known or suspected infection and indirect spatial transmission risk arising from the local microepidemiological environment.

2.11.2. Statistical Model

A multinomial logistic regression model was fitted with cluster membership as the polytomous outcome variable. Cluster 0 comprising control herds with no recorded breakdown during the study period was specified as the reference category, such that all estimated associations reflect the log-odds of belonging to a given breakdown phenotype relative to remaining breakdown-free. The model was estimated simultaneously across all three non-reference outcome categories. For $k \in \{1, 2, 3\}$, the model takes the form:

$$\begin{aligned} \log\left(\frac{P(\text{Cluster} = k \mid \mathbf{x})}{P(\text{Cluster} = 0 \mid \mathbf{x})}\right) = & \beta_{k0} + \beta_{k1} \text{Control_moves} + \beta_{k2} \text{NewBD_moves} \\ & + \beta_{k3} \text{Burnout_moves} + \beta_{k4} \text{Unknown_moves} \\ & + \beta_{k5} \text{Burnout_neighbours} + \beta_{k6} \text{Control_neighbours} \\ & + \beta_{k7} \text{NewBD_neighbours} \end{aligned} \quad (1)$$

where $\mathbf{x}$ denotes the vector of predictor variables and β_{k0} is the cluster-specific intercept. The model was implemented in R using the `multinom()` function from the `nnet` package, with a maximum of 500 optimisation iterations and convergence assessed by the relative change in the log-likelihood between successive iterations. Statistical significance was assessed using Wald z -tests, and associations are reported as odds ratios (ORs) with 95% confidence intervals (CIs) obtained by exponentiating the estimated log-odds coefficients and their standard

errors.

Model fit was assessed using the likelihood-ratio test against an intercept-only null model, with the test statistic asymptotically distributed as χ^2 under the null hypothesis of no association between predictors and cluster membership. McFadden’s pseudo- R^2 was additionally computed as a summary measure of explained variation, with the expectation that values would be modest given the large number of structurally similar control observations and the known conservatism of this index in large samples. Discriminative performance was evaluated using one-versus-rest area under the receiver operating characteristic curve (AUC) for each outcome class individually and the Hand–Till multiclass AUC, providing insight into whether predictive discrimination was concentrated between control herds and breakdown clusters or extended to meaningful separation among the three breakdown phenotypes themselves.

The robustness of estimated associations to the random selection of the 30,000 control herd records was assessed by refitting the multinomial model under three independent random seeds and comparing the resulting odds ratios across replicates; consistency of key estimates across seeds was taken as evidence that findings were not materially sensitive to the particular control sample drawn. To examine whether the association between neighbourhood infection pressure and breakdown cluster membership varied over time as national bTB prevalence fluctuated across the study period a secondary model incorporating interaction terms between `NewBD_neighbours` and `Burnout_neighbours` with calendar year was fitted. As control herds have no associated breakdown year by definition, this secondary interaction model was estimated exclusively among breakdown herds (clusters 1–3), with cluster 1 specified as the reference category, and its results are interpreted as reflecting temporal heterogeneity in the influence of local neighbourhood infection pressure on breakdown phenotype rather than on breakdown occurrence per se.

2.11.3. Survival analysis

Time-to-event methods were used to characterise cluster differences in two distinct outcomes: breakdown duration and time to subsequent breakdown.

For breakdown duration, the outcome was the total number of days from breakdown initiation to confirmed herd clearance. As the end date of each breakdown was recorded for all events in the cohort, no censoring was required for this outcome and all events were treated as complete.

For time to subsequent breakdown, the outcome was the number of days from the clearance of the index breakdown to the initiation of the next confirmed breakdown in the same herd. Herds with no recorded subsequent breakdown within the study period were right-

censored at 1,000 days, a threshold selected to capture the majority of recurrence events while limiting the distortion associated with administrative right-censoring at the study end date.

Kaplan–Meier survival curves were estimated for each cluster and differences between curves were assessed using the log-rank test. The survival function $S(t) = P(T > t)$ represents the probability that a herd has not yet experienced the event either remaining under breakdown restriction or remaining breakdown-free following clearance beyond time t . Cluster differences in time to subsequent breakdown were further examined using Cox proportional hazards regression, with adjustment for herd size (`herd_size_mean`), long-term breakdown history (`no_bds_started_in_prev_5yr`), and prior short-term recurrence (`bd_within_13_months`). This allowed assessment of whether the observed cluster differences in recurrence speed were robust to confounding by herd-level characteristics, supporting the conclusion that cluster membership captures genuine epidemiological heterogeneity beyond what is explained by herd size or history alone. The proportional hazards assumption was assessed for each covariate using Schoenfeld residuals and the `cox.zph` test.

2.11.4. Software and reproducibility

All analyses were conducted using R (version 4.5.0) within RStudio on a Windows environment, with a fixed random seed (`set.seed(123)`) to ensure reproducibility. The analysis utilised several packages, including `ClusterR` (for K-means optimisation), `factoextra` (for visualisation and cluster validation), `cluster` (for PAM clustering), `mclust` (for Gaussian Mixture Models), `dplyr` (for data manipulation), and `ggplot2` (for visualisation) [20],[21]. All code used for data processing, analysis, and visualisation is publicly available at: <https://github.com/victorychukwudi/Modeling-Cattle-TB-Spread-with-Networks>.

3. Results

3.1. Descriptive Statistics

Descriptive statistics for the five variables entered into the clustering algorithm are presented in Table 1, alongside two additional breakdown characteristics recorded for descriptive purposes. Considerable variability was observed across herds for all continuous measures. Breakdown duration ranged from 60 to 3,042 days (mean 184, SD 101), reflecting marked heterogeneity in the persistence of infection within herds. The spread ratio exhibited similarly wide dispersion (mean 0.34, SD 1.81; maximum 102), indicative of occasional but extreme within-herd amplification events. The proportion of animals testing positive at the index test was low on average (mean 0.07), yet spanned the full range from near-zero to unity, suggesting that breakdowns were detected at widely varying stages of progression.

Herd size varied substantially (mean 101, SD 110; maximum 2,231), consistent with the structural diversity of the national cattle population. Prior breakdown history was uncommon, with most herds recording no breakdowns in the preceding five years (mean 0.22, SD 0.47). Within-herd spread was observed in 18.3% of breakdowns, while recurrence within 13 months occurred in 3.6% of cases. Collectively, this variability across epidemiological, temporal, and structural dimensions provides a strong empirical basis for applying cluster analysis to identify distinct breakdown profiles.

Table 1: Descriptive statistics of clustering variables and additional breakdown characteristics

Variable	Mean	SD	Min	Max	Role
<i>Clustering variables</i>					
Proportion positive at index test	0.07	0.11	0.00	1.00	Clustering
Spread ratio	0.34	1.81	0.00	102	Clustering
Breakdown duration (days)	184	101	60	3,042	Clustering
Number of prior breakdowns (5 years)	0.22	0.47	0	6	Clustering
Mean herd size	101	110	0	2,231	Clustering
<i>Additional breakdown characteristics</i>					
Within-herd spread (any)	0.18	0.39	0	1	Descriptive
Recurrence within 13 months	0.04	0.19	0	1	Descriptive

3.2. Exploratory Data Analysis

Initial exploratory analysis revealed that several variables were strongly right-skewed, with most herds exhibiting low-severity breakdowns and a small number of high-severity outliers (Supplementary Material 1 Figure 5). Log-transformation substantially reduced skewness and stabilised variance across continuous variables, improving their suitability for distance-based clustering while preserving key epidemiological patterns (Supplementary Material 1 Figure 6).

Density plots of the transformed variables further confirmed improved distributional symmetry and reduced extreme dispersion (Supplementary Material 1 Figure 7). Examination of pairwise relationships using scatter plot matrices indicated varying degrees of association between variables, with some evidence of clustering structure and non-linear relationships (Supplementary Material 1 Figure 8).

Correlation analysis demonstrated moderate associations among selected variables, with no evidence of excessive multicollinearity that would compromise clustering performance

(Supplementary Material 1 Figure 9). Collectively, these exploratory steps supported the suitability of the transformed variables for subsequent multivariate and clustering analyses.

3.3. PCA Results

3.3.1. Structure of the PCA Space

The scree plot (Figure 14) illustrates the proportion of total variance explained by each of the five principal components. PC1 accounted for the largest share of variance (34.5%), followed by PC2 (28.5%), PC3 (19.0%), PC4 (11.0%), and PC5 (6.9%), with the first two components collectively explaining 63.0% of total variance and the first four explaining 93.1%. The relatively gradual decline across components, with no single component dominating and no sharp elbow, reflects the multidimensional nature of bTB breakdown heterogeneity, in which severity, spread, duration, recurrence, and herd size each contribute meaningfully to the overall structure of the data.

Visualisation of observations in the plane defined by the first two principal components (Supplementary Material 1 Figure 17) revealed a structured but continuous distribution without sharply delineated boundaries. A dense concentration of observations near the origin reflects the predominance of low-severity, self-limiting breakdowns in the cohort, while a broader scatter extending along PC1 and PC2 captures more extreme breakdown events characterised by elevated spread, prolonged duration, or chronic recurrence. The absence of clearly separated clusters in the two-dimensional projection is consistent with the overlapping and continuous nature of bTB breakdown dynamics, and does not preclude the identification of meaningful cluster structure by distance-based methods operating in the full five-dimensional feature space.

3.3.2. Variable Contributions to Principal Components

Table 2: Principal component loadings for the five clustering variables across the first three principal components.

Variable	PC1	PC2	PC3
prop_positive_index	0.040	0.565	0.475
spread_ratio_log	-0.553	0.133	-0.091
bd_duration_days_log	-0.539	0.118	0.070
no_bds_started_in_prev_5yr_log	-0.123	-0.421	0.533
herd_size_mean_log	-0.208	-0.567	-0.322

Examination of variable contributions to the first three principal components (Table 2; Supplementary Material 1 Figure 19) revealed distinct and epidemiologically interpretable

loading patterns, confirming that each component captured a substantively different dimension of breakdown heterogeneity.

PC1 was dominated by breakdown duration (`bd_duration_days_log`), within-herd spread ratio (`spread_ratio_log`), and herd size (`herd_size_mean_log`), collectively reflecting the acute transmission dimension of breakdown behaviour. PC1 represents a gradient of outbreak severity and persistence, separating self-limiting low-intensity breakdowns from explosive, difficult-to-clear events consistent with the Cluster 3 phenotype.

PC2 was driven principally by herd size (`herd_size_mean_log`), index test positivity (`prop_positive_index`), and breakdown duration (`bd_duration_days_log`), capturing variation in herd scale and apparent prevalence at the point of detection, independently of subsequent within-herd spread.

PC3 was defined predominantly by prior breakdown history (`no_bds_started_in_prev_5yr_log`), capturing the recurrence dimension orthogonally to the acute transmission and herd-scale dimensions represented by PC1 and PC2. The separation of the recurrence signal onto a third independent component confirms that chronic recurrence represents a genuinely distinct epidemiological dimension, reinforcing the validity of Cluster 2 as a mechanistically separate breakdown phenotype.

Together, the loading patterns across PC1, PC2, and PC3 confirm that the five clustering variables capture three largely independent dimensions of bTB breakdown heterogeneity: acute transmission severity and persistence, herd scale and index prevalence, and chronic recurrence history providing a multivariate epidemiological basis for the three-cluster PAM solution.

3.4. Clustering Results

3.4.1. Cluster number selection

The optimal number of clusters was determined by convergent evidence across four independent criteria: the within-cluster sum of squares elbow method, the variance-explained elbow method, the average silhouette width, and majority vote across 26 validity indices implemented in `NbClust`. Across these criteria, $K = 3$ received the strongest and most consistent support. While a small number of indices favoured $K = 4$ or $K = 5$, their support was substantially weaker and inconsistent across methods. The gap statistic suggested $K = 1$, which is a known artefact of this method in large datasets with overlapping cluster boundaries, and was treated as supplementary evidence only. This multi-criterion convergence on $K = 3$ provides robust justification for the selected solution independently of any single index.

Average silhouette widths for PAM across $K = 2$ to $K = 4$ were shown in (Supplemen-

tary Material 1 Figure 13). The silhouette width was maximised at $K = 3$ (0.391), with lower values observed at $K = 2$ (0.362) and $K = 4$ (0.311). The monotonic decline in silhouette width for $K \geq 4$ indicates that additional clusters partition existing structure rather than reveal substantively distinct groupings, with increasing within-cluster heterogeneity and inter-cluster overlap. The $K = 3$ solution therefore represents the most parsimonious configuration that maximises both cluster cohesion and separation within the PAM framework.

3.4.2. Bootstrap Stability

The stability of the PAM $K = 3$ solution was assessed using bootstrap resampling via the `clusterboot` function, in which the full dataset was resampled with replacement across $B = 200$ iterations and PAM was re-applied in each iteration. The mean Jaccard coefficient between bootstrapped and canonical cluster assignments was computed per cluster, with values exceeding 0.85 considered indicative of a stable cluster and values below 0.60 indicative of an unstable or dissolved cluster.

All three clusters demonstrated outstanding stability across the 200 bootstrap resamples. Mean Jaccard coefficients were 0.996 for Cluster 1, 0.998 for Cluster 2, and 0.982 for Cluster 3, with no cluster dissolved in any of the 200 resamples (Times Dissolved = 0 for all clusters). All values substantially and uniformly exceeded the 0.85 threshold for stable cluster classification, indicating that the three breakdown phenotypes were recovered with near-perfect consistency across repeated resampling of the data and confirming that the cluster solution reflects genuine and reproducible structure in the underlying breakdown population rather than an artefact of the particular observations on which PAM was fitted.

The exceptionally high Jaccard coefficients observed across all three clusters, combined with zero dissolution events across 200 resamples, provide strong evidence that the three-cluster PAM solution is robust to sampling variation. The stability of Cluster 2 (0.998) is particularly noteworthy: this phenotype is defined principally by elevated prior breakdown history a variable with a highly skewed distribution concentrated at zero yet the cluster was recovered with near-perfect fidelity across all bootstrap resamples, indicating that the recurrence signal driving this phenotype is sufficiently strong and consistent to withstand repeated perturbation of the input data. Cluster 3, which captures the acute high-intensity transmission phenotype characterised by elevated spread ratio and prolonged breakdown duration, achieved a Jaccard coefficient of 0.982 despite exhibiting the greatest within-cluster heterogeneity in breakdown duration, further confirming the structural robustness of this grouping. These findings collectively satisfy a critical criterion for the validity of an unsupervised classification intended for epidemiological inference: that the identified phenotypes

are stable, reproducible, and independent of the specific observations included in any single run of the clustering algorithm.

3.4.3. Split-Sample Validation

To assess whether the PAM $K = 3$ solution generalises beyond the specific observations used to fit it, split-sample validation was performed across five independent random partitions of the data. In each replicate, 70% of observations were randomly assigned to a training partition on which PAM was re-fitted, and the remaining 30% were assigned to a held-out test partition. Cluster membership for test observations was predicted by nearest-medoid assignment, and agreement between predicted and canonical cluster labels was quantified using the Adjusted Rand Index (ARI), where $\text{ARI} = 1$ denotes perfect agreement and $\text{ARI} = 0$ denotes agreement at the level expected by chance .

The results demonstrated near-perfect agreement between the training-derived and canonical cluster assignments across all five splits, with ARI values ranging from 0.995 to 1.000 and a mean ARI of 0.999 (SD = 0.002). Three of the five splits achieved perfect agreement (ARI = 1.000), indicating exact recovery of the canonical cluster solution on unseen data. The minimum ARI of 0.995 in Split 2 reflects negligible misclassification, confirming that the cluster solution is not sensitive to the particular observations included in the fitting partition.

These results provide compelling evidence of the generalisability and structural robustness of the three-cluster PAM solution. The near-perfect recovery of cluster assignments on held-out data across all five replicates confirms that the breakdown phenotypes identified by PAM are not an overfitting artefact of the full dataset but represent stable, reproducible epidemiological groupings that are consistently recoverable from any representative subsample of the data.

3.4.4. Sequential Variable Removal

To assess the influence of individual clustering variables on the overall solution and identify whether any single variable dominated the cluster structure, each of the five variables was removed in turn and PAM $K = 3$ was re-applied to the reduced four-variable standardised feature matrix. Agreement between the reduced and canonical solutions was quantified by the ARI.

The results revealed a clear distinction between variables that were structurally influential and those that contributed supplementary information. Removal of `spread_ratio_log` (ARI = 0.328), `bd_duration_days_log` (ARI = 0.326), and `no_bds_started_in_prev_5yr_log` (ARI = 0.273) each produced large reductions in agreement with the canonical solution,

indicating that these three variables were the primary structural drivers of the cluster solution corresponding respectively to the within-herd transmission, breakdown persistence, and chronic recurrence dimensions that define the three breakdown phenotypes. In contrast, removal of `prop_positive_index` (ARI = 0.972) and `herd_size_mean_log` (ARI = 0.934) produced minimal disruption to the cluster structure, confirming that index test positivity and herd size contribute supplementary rather than defining information to the solution. Importantly, no single variable alone accounted for the entire cluster structure, confirming that the solution reflects genuine multivariate epidemiological heterogeneity rather than the dominance of one variable.

3.4.5. Description of Clusters

Mean values of the five clustering variables within each PAM cluster, reported on their original untransformed scales, are presented in Table 3. The three clusters exhibited clear, epidemiologically interpretable, and mutually distinct profiles across all five dimensions, confirming that the PAM solution captures genuine heterogeneity in breakdown dynamics rather than arbitrary partitions of a homogeneous population.

Table 3: Mean values of the five clustering variables by PAM cluster ($K = 3$), reported on the original untransformed scale. Cluster sizes are given in parentheses.

Cluster	Index positivity	Spread ratio	Duration (days)	Prior BDs (5 yr)	Herd size
Cluster 1 ($n = 25,940$)	0.073	0.004	148.9	0.000	85.7
Cluster 2 ($n = 7,227$)	0.053	0.073	177.4	1.109	130.5
Cluster 3 ($n = 6,506$)	0.073	1.975	333.0	0.126	130.6

Cluster 1 — Low-intensity, self-limiting breakdowns ($n = 25,940$; 65.8% of the cohort). Cluster 1 comprised the substantial majority of breakdown events and displayed uniformly low values across all five clustering dimensions. Within-herd spread was negligible (mean spread ratio 0.004), indicating that virtually no additional reactors were identified beyond the index test in the typical Cluster 1 breakdown. Mean breakdown duration was the shortest of the three phenotypes at 149 days, prior breakdown history was effectively absent (mean 0.000 breakdowns in the preceding five years), and mean herd size was the smallest (86 animals). Index test positivity (7.3%) was comparable to Cluster 3 but substantially higher than Cluster 2. This profile is consistent with a low-intensity, self-limiting breakdown phenotype in which infection is detected at a relatively early stage and resolved rapidly

with minimal within-herd amplification, representing the least epidemiologically complex and least resource-intensive breakdown type in the cohort.

Cluster 2 — Chronic recurrence without active spread ($n = 7,227$; 18.3% of the cohort). Cluster 2 was defined by a markedly elevated prior breakdown history relative to both other clusters, with a mean of 1.11 breakdowns initiated in the preceding five years more than 8.8 times greater than Cluster 3 (0.126) and effectively infinite relative to Cluster 1 (0.000). Despite this chronic recurrence profile, index test positivity was the lowest of the three clusters (5.3%), within-herd spread was limited (mean spread ratio 0.073), and breakdown duration was intermediate at 177 days. Mean herd size (130.5 animals) was substantially larger than Cluster 1, suggesting that herd scale may contribute to persistent exposure risk. The combination of frequent recurrence in the absence of active within-herd transmission is inconsistent with a model of repeated de novo introduction and explosive spread, and instead points to a pattern of residual infection persisting below the detection threshold of routine skin testing, or continual low-level re-exposure from a persistent environmental or wildlife source. These herds may be characterised as epidemiologically “quiet” in any individual breakdown but chronically and repeatedly at risk a phenotype that poses a particular challenge for eradication programmes reliant on test-and-removal as the primary clearance mechanism.

Cluster 3 — Acute high-intensity transmission in naïve herds ($n = 6,506$; 16.5% of the cohort). Cluster 3 exhibited the most intensive acute breakdown profile of the three phenotypes across all transmission-related variables. The mean spread ratio of 1.975 indicates that, on average, almost twice as many reactor animals were identified at subsequent testing rounds as at the index test, reflecting sustained within-herd amplification of infection across successive rounds of testing. Mean breakdown duration was the longest at 333 days more than twice that of Cluster 1 and 88% longer than Cluster 2 consistent with the interpretation that repeated testing cycles required to identify and remove additional reactors substantially extend the time to clearance. Index test positivity (7.3%) was comparable to Cluster 1. In marked contrast to Cluster 2, prior breakdown history was low (mean 0.126 breakdowns in the preceding five years), suggesting that Cluster 3 herds were largely epidemiologically naïve to bTB infection at the time of the index event. This phenotype is consistent with a susceptible herd encountering bTB infection for the first time whether through an infected animal introduction or intense localised transmission pressure from the surrounding landscape and experiencing an explosive, difficult-to-clear outbreak driven by sustained within-herd spread before ultimately achieving clearance.

Taken together, the three phenotypes reflect fundamentally distinct underlying epidemiological mechanisms: rapid detection and resolution of low-level infection (Cluster 1), chronic

recurrence driven by persistent background exposure in the absence of active within-herd spread (Cluster 2), and acute high-intensity transmission in epidemiologically naïve herds (Cluster 3). The K -means and PAM solutions produced broadly consistent cluster structures and near-identical mean profiles, with the principal methodological difference being that PAM resolved the recurrence dimension characterising Cluster 2 more distinctly, consistent with its known robustness to the influence of individual variables with extreme distributional profiles. This taxonomy provides a novel construct for understanding bTB breakdown heterogeneity that moves beyond aggregate population-level descriptions toward a mechanistic characterisation of distinct infection dynamics, with direct implications for the design of targeted and proportionate disease management interventions.

3.5. Temporal distribution of clusters

The temporal distribution of clusters was examined by calculating the annual proportion of herds assigned to each PAM cluster ($K = 3$). The results are summarised in (Table 4) and illustrated in (Supplementary Material 1 Figure 25).

Table 4: Annual proportion of herds assigned to each cluster.

Year	Cluster 1	Cluster 2	Cluster 3
2007	0.58	0.15	0.26
2008	0.66	0.10	0.23
2009	0.64	0.16	0.19
2010	0.61	0.20	0.19
2011	0.62	0.18	0.20
2012	0.62	0.19	0.20
2013	0.65	0.17	0.18
2014	0.68	0.15	0.17
2015	0.68	0.17	0.15
2016	0.68	0.17	0.16
2017	0.68	0.14	0.18
2018	0.69	0.15	0.16
2019	0.67	0.15	0.18
2020	0.66	0.16	0.19
2021	0.67	0.15	0.17
2022	0.64	0.17	0.19
2023	0.64	0.18	0.19
2024	0.75	0.17	0.09

Cluster 1 consistently represents the majority of herds across all years, with proportions generally ranging between approximately 0.61 and 0.69, indicating a stable dominant profile over time. Clusters 2 and 3 remain smaller and relatively stable, each typically accounting for between 0.14 and 0.20 of observations.

While minor year-to-year fluctuations are observed, no substantial structural shifts in cluster composition are evident. A notable deviation occurs in 2024, where Cluster 1 increases markedly (approximately 0.75), accompanied by a reduction in Cluster 3. This pattern should be interpreted cautiously, as it may reflect incomplete or partially observed data for the most recent year.

Collectively, the stability of cluster proportions suggests that the clustering captures persistent underlying epidemiological structure rather than short-term variation.

3.6. Multinomial Regression Results

The multinomial logistic regression model was fitted to a dataset of 66,688 observations comprising 30,000 control herds (Cluster 0) and 36,688 breakdown herds distributed across three PAM-derived phenotype clusters. The model converged successfully and provided a significantly better fit than the intercept-only null model (likelihood ratio test: $\chi^2 = 10,797$, $df = 21$, $p < 2.2 \times 10^{-16}$), confirming that movement and neighbourhood exposure variables collectively discriminate between control herds and the three breakdown phenotypes. McFadden’s pseudo- R^2 was 0.068, which, while modest in absolute terms, is consistent with expectations for a large multinomial model with four outcome categories and reflects the inherent difficulty of predicting specific breakdown phenotype from population-level exposure metrics alone. The residual deviance was 148,134.6 (AIC = 148,182.6).

Average predicted probabilities across the analytical dataset were 0.450 for control (Cluster 0), 0.346 for Cluster 1, 0.096 for Cluster 2, and 0.108 for Cluster 3, broadly consistent with the observed cluster proportions and confirming adequate model calibration across all four outcome categories.

3.7. Odds Ratios and Confidence Intervals

Odds ratios, 95% confidence intervals, and p -values for each predictor are presented in Table 5, with all comparisons expressed relative to control herds (Cluster 0). Odds ratio plots for neighbourhood and movement predictors are presented in Supplementary Material 1 Figures 26 and 27 respectively.

Table 5: Odds ratios (OR) and 95% confidence intervals from the multinomial logistic regression model predicting breakdown cluster membership relative to control herds (Cluster 0). Confidence intervals computed from Wald standard errors. p -values from two-tailed Wald z -tests.

Predictor	Cluster 1 vs. 0			Cluster 2 vs. 0			Cluster 3 vs. 0		
	OR	95% CI	p	OR	95% CI	p	OR	95% CI	p
<i>Movement variables</i>									
Control_moves	0.998	(0.998, 0.999)	<0.001	1.000	(1.000, 1.001)	0.769	0.998	(0.997, 0.999)	<0.001
NewBD_moves	1.003	(0.999, 1.006)	0.128	1.003	(0.999, 1.007)	0.176	1.005	(1.001, 1.009)	0.020
Burnout_moves	0.999	(0.997, 1.001)	0.379	0.999	(0.997, 1.001)	0.411	1.004	(1.002, 1.006)	<0.001
Unknown_moves	1.000	(0.979, 1.022)	0.993	1.032	(1.010, 1.054)	0.004	1.010	(0.983, 1.037)	0.472
<i>Neighbourhood variables</i>									
Burnout_neighbours	1.024	(1.020, 1.028)	<0.001	1.048	(1.043, 1.053)	<0.001	1.041	(1.037, 1.046)	<0.001
Control_neighbours	0.987	(0.986, 0.988)	<0.001	0.985	(0.984, 0.986)	<0.001	0.983	(0.982, 0.984)	<0.001
NewBD_neighbours	1.372	(1.362, 1.382)	<0.001	1.456	(1.444, 1.468)	<0.001	1.473	(1.461, 1.485)	<0.001

95% confidence intervals computed from Wald standard errors. Cluster 0 = control herds (reference

category). p -values from two-tailed Wald z -tests.

Neighbourhood infection pressure variables produced consistently larger and more statistically robust associations with breakdown cluster membership than inward movement variables across all three phenotypes. **NewBD_neighbours** was the strongest predictor in the model, with odds ratios of 1.372 (Cluster 1), 1.456 (Cluster 2), and 1.473 (Cluster 3), all highly significant ($p < 0.0001$), indicating that each additional new breakdown neighbour was associated with a 37–47% increase in the odds of belonging to a breakdown phenotype relative to remaining breakdown-free. The gradient in odds ratios across clusters weakest for Cluster 1 and strongest for Cluster 3 suggests that neighbourhood infection pressure not only increases the probability of breakdown occurrence but preferentially drives membership in the more severe, acute high-intensity transmission phenotype.

Burnout_neighbours was similarly significant across all clusters (ORs: 1.024, 1.048, 1.041; all $p < 0.0001$), consistent with ongoing or resolving breakdowns constituting a source of residual environmental and contact-based transmission pressure. **Control_neighbours** exhibited a consistent protective association across all three clusters (ORs: 0.987, 0.985, 0.983; all $p < 0.0001$), reflecting reduced breakdown risk for herds embedded within a predominantly unaffected neighbourhood.

Among movement variables, **Burnout_moves** reached statistical significance for Cluster 3 only (OR = 1.004, $p < 0.001$), and **NewBD_moves** was significant only for Cluster 3 (OR = 1.005, $p = 0.020$). **Control_moves** was non-significant for Clusters 1 and 3, with a borderline significant negative association for Cluster 1 (OR = 0.998, $p < 0.001$). **Unknown_moves** was significant only for Cluster 2 (OR = 1.032, $p = 0.004$). The consistently small effect sizes

for movement variables (ORs close to 1.00) relative to the substantially larger effects of neighbourhood variables confirms the dominance of local spatial transmission pressure over direct trade-mediated exposure as a determinant of breakdown cluster membership.

3.7.1. Robustness to Control Sampling

To assess whether findings were sensitive to the random selection of the 30,000 control herd records, the multinomial model was refitted under three independent random seeds. Odds ratios for `NewBD_neighbours` ranged from 1.352 to 1.359 for Cluster 1, 1.431 to 1.439 for Cluster 2, and 1.451 to 1.459 for Cluster 3 across seeds demonstrating negligible sensitivity to the particular control sample drawn. All neighbourhood predictors remained highly significant ($p < 0.0001$) across all three seeds, confirming that the key findings are robust to random variation in control herd selection.

3.7.2. Temporal Variation in Neighbourhood Effects

To examine whether the association between neighbourhood infection pressure and breakdown cluster membership varied over time, a secondary multinomial model was fitted among breakdown herds only (Clusters 1–3, $n = 36,688$), incorporating interaction terms between `NewBD_neighbours` and `Burnout_neighbours` with calendar year. Control herds were excluded as they have no associated breakdown year.

Both interaction terms were highly significant for all cluster comparisons ($p < 0.0001$), confirming that the association between neighbourhood infection pressure and breakdown phenotype varied systematically over time. The interaction of `NewBD_neighbours` with calendar year yielded odds ratios of 1.010 (Cluster 2 vs 1) and 1.004 (Cluster 3 vs 1) per unit increase in year, indicating a gradual strengthening of the neighbourhood new breakdown effect on cluster membership over the study period. Conversely, the interaction of `Burnout_neighbours` with calendar year yielded odds ratios of 0.992 (Cluster 2 vs 1) and 0.991 (Cluster 3 vs 1), suggesting a modest attenuation of the burnout neighbourhood effect over time. These temporal dynamics are consistent with changing patterns of national bTB prevalence across the study period and warrant further investigation in analyses with extended follow-up.

3.7.3. Model Discrimination

One-versus-rest AUC values were 0.723 for Cluster 0 (control herds), 0.611 for Cluster 1, 0.673 for Cluster 2, and 0.671 for Cluster 3. The Hand–Till multiclass AUC was 0.626, indicating moderate overall discrimination across all four outcome categories. The highest one-versus-rest AUC for Cluster 0 confirms that the model discriminates most effectively between herds with no breakdown history and those experiencing any breakdown phenotype.

The comparable AUC values for Clusters 2 and 3 (0.673 and 0.671) indicate that the model achieves meaningful discrimination between the chronic recurrence and acute high-intensity phenotypes relative to the remaining classes, supporting the epidemiological relevance of the neighbourhood and movement predictors beyond simple breakdown versus no-breakdown classification.

3.8. Marginal Effects of Risk Factor Predictors

To illustrate the practical magnitude of associations identified in the multinomial model, predicted probabilities of cluster membership were computed at the 10th, 50th, and 90th percentiles of each predictor, holding all other variables at their observed means.

Neighbourhood infection pressure exerted the largest and most epidemiologically meaningful effects on predicted cluster membership probabilities, substantially exceeding the influence of inward movement variables across all three breakdown phenotypes.

NewBD_neighbours produced the most pronounced gradient of any predictor. At the 10th percentile, the predicted probability of remaining breakdown-free was 63.1%, declining sharply to 18.2% at the 90th percentile. Correspondingly, the predicted probability of belonging to Cluster 1 increased from 25.3% to 48.7%, Cluster 2 from 5.6% to 15.3%, and Cluster 3 from 6.1% to 17.8% across the same range. This near-monotonic shift from breakdown-free status toward all three breakdown phenotypes with increasing new breakdown neighbourhood pressure provides strong evidence that local microepidemic dynamics driven by the presence of actively breaking-down herds in the surrounding area are a dominant determinant of both breakdown occurrence and breakdown phenotype.

Control_neighbours exhibited the opposite pattern, consistent with a protective effect of being surrounded by unaffected herds. At the 10th percentile, the predicted probability of remaining breakdown-free was 35.6%, rising to 51.7% at the 90th percentile, while predicted probabilities of belonging to Clusters 1, 2, and 3 declined from 40.9%, 10.9%, and 12.6% to 32.2%, 7.8%, and 8.3% respectively.

Burnout_neighbours produced a similar but smaller gradient, with breakdown-free probability declining from 46.2% at the 10th percentile to 37.2% at the 90th percentile, and predicted probabilities of Clusters 2 and 3 increasing from 8.2% and 9.5% to 11.6% and 12.4% respectively. The smaller magnitude relative to **NewBD_neighbours** is consistent with the interpretation that actively breaking-down herds represent a more immediate transmission risk than those in a post-restriction recovery phase.

Inward movement variables (**Control_moves**, **NewBD_moves**, **Burnout_moves**, **Unknown_moves**) produced minimal gradients across their observed ranges, with predicted probabilities varying by less than two percentage points between the 10th and 90th percentiles for all four

predictors. **Unknown_moves** produced identical predicted probabilities at all three percentile levels, indicating negligible independent discriminative information once neighbourhood effects were accounted for.

Taken together, these marginal effect estimates confirm a clear asymmetry between the two domains of epidemiological exposure: neighbourhood infection pressure reflecting the local epidemiological context in which a herd operates is a substantially stronger determinant of breakdown phenotype than the infection status of inward cattle movements. This finding supports a local microepidemic mechanism in which spatially clustered transmission, whether mediated through shared wildlife reservoirs, environmental contamination, or undetected inter-herd contact, drives both the occurrence and the character of bTB breakdowns more powerfully than direct trade-related exposure.

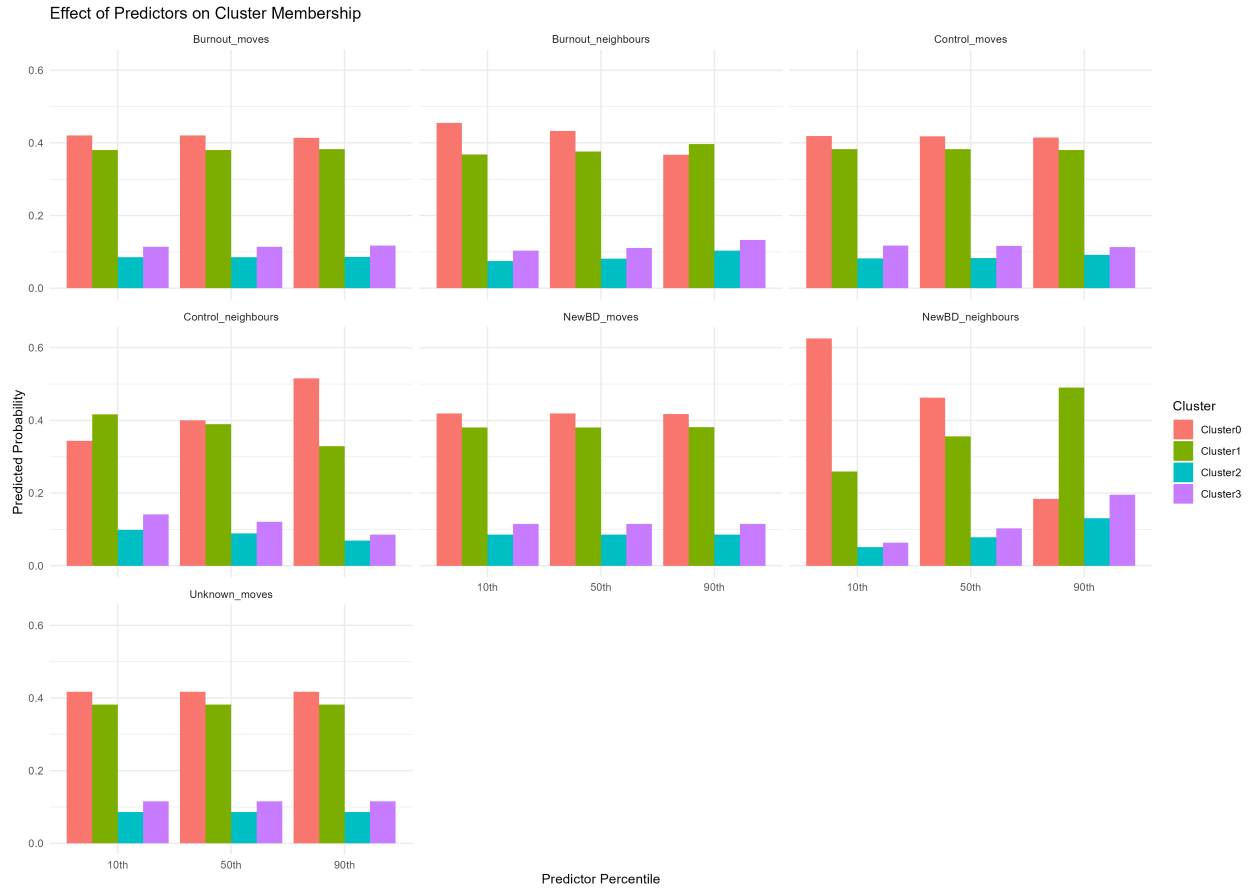


Figure 1: Predicted probabilities of cluster membership at the 10th, 50th, and 90th percentiles of each risk factor predictor, with all other predictors held at their observed means. Neighbourhood variables particularly **NewBD_neighbours** and **Control_neighbours** show the largest gradients, confirming their dominance over movement variables as determinants of breakdown cluster membership.

4. Survival Analysis Results

4.1. Breakdown Duration

Kaplan–Meier curves were estimated to describe the probability of a breakdown remaining active over time, stratified by cluster. All breakdowns ultimately resolved, as all events were treated as complete; however, the rate of clearance differed markedly across clusters. Clusters 1 and 2 exhibited rapid and near-identical declines in the probability of remaining in breakdown, with the majority of herds achieving clearance within approximately 300 days of breakdown initiation. The close correspondence between Clusters 1 and 2 in this analysis is consistent with their similar mean breakdown durations (149 and 177 days respectively; Table 3), reflecting that despite their distinct recurrence profiles, both phenotypes resolve at comparable rates once a breakdown has been initiated.

Cluster 3, by contrast, exhibited a substantially prolonged breakdown duration, with the survival curve declining at a markedly slower rate than both other clusters. A substantial proportion of Cluster 3 herds remained in breakdown beyond 500 days, and the risk table confirms that 53 herds remained active at 1,000 days, four at 2,000 days, and one at 3,000 days reflecting extreme persistence in a subset of the most severely affected herds. This is consistent with the mean breakdown duration of 333 days observed for Cluster 3 and with its high spread ratio, as repeated rounds of testing to identify and remove additional reactors necessarily extend the time to clearance. The log-rank test confirmed highly significant differences in breakdown duration across clusters ($p < 0.0001$), providing strong statistical evidence that the three phenotypes differ not only in their epidemiological profiles but in the time burden they impose on affected herds and the national control programme.

4.2. Time to Subsequent Breakdown

Kaplan–Meier curves describing the probability of a herd remaining breakdown-free following clearance of the index breakdown are presented in Figure 2. The relatively gradual overall decline of the curves reflects that the majority of herds did not experience a subsequent breakdown within the 1,000-day observation window, indicating that recurrence, while meaningful, is not universal. The log-rank test demonstrated a highly statistically significant difference in time to subsequent breakdown across clusters ($p < 0.0001$), confirming that recurrence dynamics vary substantially across the identified phenotypes.

Cluster 1 exhibited the slowest decline in breakdown-free probability, indicating that herds in this group remained clear for the longest periods following resolution of the index breakdown and experienced the lowest rate of recurrence. This is consistent with the near-zero prior breakdown history of Cluster 1 herds (Table 3), suggesting that low underlying exposure risk confers protection against rapid reinfection. Cluster 2 exhibited the most rapid

decline in breakdown-free survival probability, with a steeper early trajectory than both other clusters, reflecting a substantially higher likelihood of early subsequent breakdown. This finding is particularly notable given that Cluster 2 herds showed limited within-herd spread during the index breakdown, suggesting that rapid recurrence in this group is driven by persistent environmental or wildlife exposure rather than residual infection amplified during the index event. Cluster 3 displayed an intermediate pattern, declining more slowly than Cluster 2 but more rapidly than Cluster 1, consistent with the view that the high-intensity acute transmission characteristic of this phenotype is not necessarily accompanied by chronically elevated recurrence risk once clearance has been achieved.

Taken together, the survival analyses demonstrate that the three breakdown phenotypes identified by PAM clustering differ not only in their within-breakdown epidemiological characteristics but in their longitudinal trajectories: the duration of restriction imposed on affected herds, and the probability and timing of subsequent disease events. These findings reinforce the epidemiological and operational relevance of phenotype-based classification as a basis for differentiated disease management.

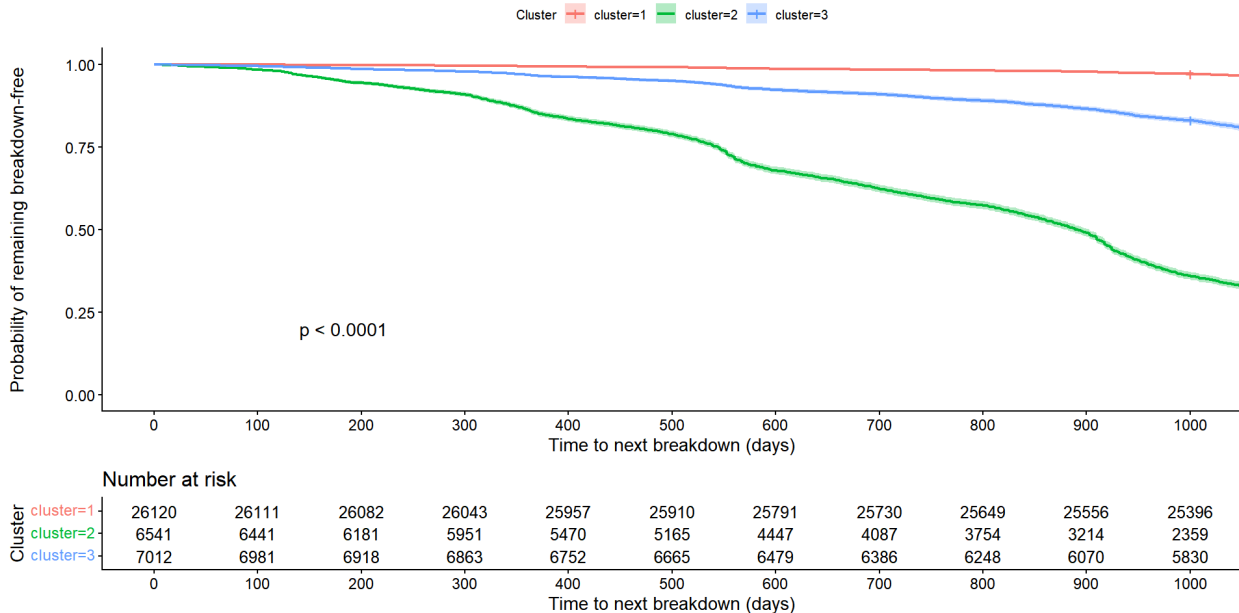


Figure 2: Kaplan–Meier curves illustrating the probability of remaining breakdown-free following clearance of the index breakdown, stratified by cluster. The y -axis represents the probability that a herd has not experienced a subsequent breakdown by each time point. Herds with no recorded subsequent breakdown within the study period were right-censored at 1,000 days. Shaded bands denote 95% confidence intervals. The risk table reports the number of herds remaining under observation at selected time points. The log-rank test confirmed a statistically significant difference in time to subsequent breakdown across clusters ($p < 0.0001$).

4.3. Cox Proportional Hazards Models

4.3.1. Breakdown Duration

A Cox proportional hazards model was fitted to examine whether cluster differences in breakdown duration persisted after adjustment for herd size, prior breakdown history, and short-term recurrence. Cluster 1 was specified as the reference category. Hazard ratios (HRs) greater than 1 indicate a faster rate of clearance (shorter duration), as the event of interest is breakdown resolution.

Table 6: Cox proportional hazards model for breakdown duration. Hazard ratios (HR) greater than 1 indicate faster clearance relative to Cluster 1. Model fitted to $n = 39,673$ breakdown events. Concordance = 0.681.

Variable	HR	95% CI	z	p
Cluster 2	0.848	(0.789, 0.911)	-4.48	<0.001
Cluster 3	0.114	(0.110, 0.118)	-120.30	<0.001
log(herd size)	0.934	(0.925, 0.943)	-14.17	<0.001
log(prior BDs)	0.541	(0.493, 0.593)	-12.97	<0.001
Short-term recurrence	0.658	(0.621, 0.697)	-14.37	<0.001

Both cluster effects were highly significant after adjustment (Table 6). Cluster 2 herds cleared their breakdowns at a modestly slower rate than Cluster 1 herds (HR = 0.848, 95% CI: 0.789–0.911, $p < 0.001$), consistent with their intermediate breakdown duration (mean 177 days versus 149 days for Cluster 1). Cluster 3 herds exhibited a markedly slower clearance rate relative to Cluster 1 (HR = 0.114, 95% CI: 0.110–0.118, $p < 0.0001$), indicating that Cluster 3 herds took substantially longer to achieve clearance even after accounting for herd size, prior breakdown history, and prior recurrence. This is consistent with the sustained within-herd transmission characteristic of the acute high-intensity phenotype, in which repeated testing rounds are required to identify and remove additional reactors. Larger herd size (HR = 0.934, $p < 0.0001$) and greater prior breakdown history (HR = 0.541, $p < 0.0001$) were both independently associated with slower clearance, as were herds with prior short-term recurrence (HR = 0.658, $p < 0.0001$).

The proportional hazards assumption was assessed using scaled Schoenfeld residuals and the `cox.zph` global test (Supplementary Material 1, Section 7, Table 8). The global test was highly significant ($\chi^2 = 2,116.4$, $df = 5$, $p < 0.0001$), indicating that the proportional hazards assumption was not met for all covariates over the full follow-up period. This is not unexpected given the extremely long tail in Cluster 3 breakdown duration (up to 3,042 days), which means that hazard ratios are not constant over time. Results should therefore

be interpreted as average effects over the observed follow-up period rather than as time-constant hazard ratios.

4.3.2. Time to Subsequent Breakdown

A Cox proportional hazards model was additionally fitted to examine cluster differences in time to subsequent breakdown after adjustment for herd size and prior breakdown history. `bd_within_13_months` was included as a covariate in the model specification but was aliased due to collinearity with prior breakdown history among herds with no subsequent breakdown event and was therefore excluded from the final model estimates. Cluster 1 was specified as the reference category, and herds with no subsequent breakdown within the study period were censored at 1,000 days ($n = 39,673$ observations; $n = 18,675$ events).

Table 7: Cox proportional hazards model for time to subsequent breakdown. Hazard ratios (HR) greater than 1 indicate faster recurrence relative to Cluster 1. Model fitted to $n = 39,673$ observations with $n = 18,675$ events. Concordance = 0.842.

Variable	HR	95% CI	z	p
Cluster 2	2.370	(2.177, 2.580)	19.89	<0.0001
Cluster 3	1.339	(1.282, 1.399)	13.16	<0.0001
log(herd size)	1.051	(1.036, 1.067)	6.60	<0.0001
log(prior BDs)	21.131	(18.959, 23.552)	55.12	<0.0001

Cluster differences in time to subsequent breakdown were highly significant after adjustment (Table 9). Cluster 2 herds experienced subsequent breakdowns at more than twice the rate of Cluster 1 herds (HR = 2.370, 95% CI: 2.177–2.580, $p < 0.0001$), confirming that the chronic recurrence phenotype is characterised by substantially accelerated rebreakdown even after controlling for herd size and prior breakdown history. This is a particularly important finding: the excess recurrence risk of Cluster 2 herds cannot be explained by their prior breakdown history alone, since that variable is already included as a covariate in the model, suggesting that additional unmeasured persistent risk factors such as environmental contamination, wildlife pressure, or structural biosecurity deficits drive the recurrence pattern of this phenotype. Cluster 3 herds also experienced subsequent breakdowns at a significantly higher rate than Cluster 1 (HR = 1.339, 95% CI: 1.282–1.399, $p < 0.0001$), reflecting an intermediate recurrence risk consistent with the Kaplan–Meier analysis. Prior breakdown history was by far the strongest predictor of recurrence (HR = 21.131, 95% CI: 18.959–23.552, $p < 0.0001$), indicating that herds with a history of repeated breakdowns face a dramatically elevated hazard of subsequent breakdown regardless of cluster membership. Larger herd size was additionally associated with modestly increased recurrence risk (HR

$= 1.051$, $p < 0.0001$).

The proportional hazards assumption was assessed using scaled Schoenfeld residuals and the `cox.zph` global test (Supplementary Material 1, Section 7, Table 9). The global test was highly significant ($\chi^2 = 857$, $df = 4$, $p < 0.0001$), indicating non-proportional hazards. Given the heterogeneous recurrence trajectories across clusters and the wide distribution of follow-up times, this finding is expected and consistent with the Kaplan–Meier curves showing crossing or diverging survival functions at different time points. Results are interpreted as average effects over the observation period.

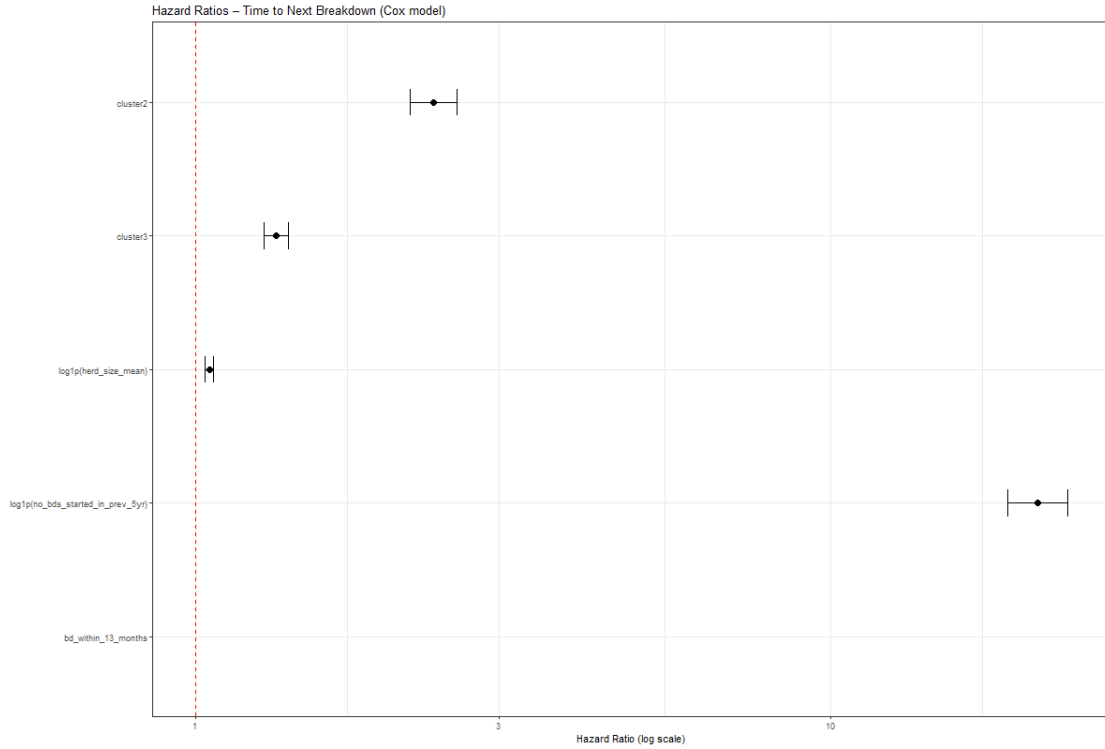


Figure 3: Forest plot of hazard ratios from the Cox proportional hazards model for time to subsequent breakdown. Hazard ratios are presented on a log scale with 95% confidence intervals. Prior breakdown history (`no_bds_started_in_prev_5yr`) is the dominant predictor of recurrence risk, followed by cluster membership. Both Cluster 2 and Cluster 3 exhibit significantly elevated recurrence hazard relative to Cluster 1 after adjustment.

Taken together, the Cox proportional hazards models confirm that the cluster differences in both breakdown duration and time to subsequent breakdown observed in the Kaplan–Meier analysis are robust to adjustment for herd size, prior breakdown history, and recurrence. The exceptionally high hazard ratio for prior breakdown history in the recurrence model ($HR = 21.131$) underscores the central role of chronic herd-level susceptibility in driving the recurrence pattern of Cluster 2, while the markedly slower clearance of Cluster 3 ($HR = 0.114$ for breakdown duration) confirms the sustained transmission burden imposed by the

acute high-intensity phenotype on affected herds and the national control programme. These findings reinforce the case for phenotype-informed intervention: Cluster 2 herds require enhanced investigation of persistent environmental and wildlife risk factors driving repeated rebreakdown, while Cluster 3 herds require intensified short-term management to accelerate clearance and limit within-herd amplification.

5. Discussion

5.1. Principal Findings

This study applied unsupervised machine learning to national bTB surveillance data to identify three epidemiologically distinct breakdown phenotypes, characterise their risk factor profiles, and quantify their longitudinal trajectories. The identification of stable, reproducible breakdown phenotypes with fundamentally different epidemiological mechanisms represents a novel contribution to the understanding of bTB heterogeneity in Irish cattle populations and provides a data-driven basis for phenotype-informed disease management.

5.2. Cluster Phenotypes and Their Epidemiological Interpretation

The three-cluster PAM solution identified breakdown phenotypes that align closely with distinct hypothesised mechanisms of bTB persistence and transmission. Cluster 1, comprising the majority of breakdown events (65.8%), represents the modal self-limiting phenotype in which infection is detected and resolved rapidly with minimal within-herd amplification. This profile is consistent with early detection through routine surveillance, effective test-and-removal, and low underlying exposure pressure. Cluster 2, characterised by chronic recurrence in the absence of active within-herd spread, is consistent with either residual infection persisting below the detection threshold of routine skin testing or continual low-level re-exposure from a persistent environmental or wildlife source. The markedly elevated hazard of subsequent breakdown in Cluster 2 herds ($HR = 2.370$ relative to Cluster 1 after full adjustment) confirms that this phenotype represents a chronically at-risk population that is not adequately addressed by standard test-and-removal protocols. Cluster 3, characterised by explosive within-herd transmission and prolonged breakdown duration in epidemiologically naïve herds, is consistent with a susceptible herd encountering bTB infection for the first time whether through an infected animal introduction or intense localised transmission pressure and experiencing sustained within-herd amplification before clearance is achieved. The markedly slower clearance rate of Cluster 3 herds ($HR = 0.114$ for breakdown duration relative to Cluster 1) reflects the repeated testing cycles required to identify and remove additional reactors generated by ongoing transmission.

These three phenotypes self-limiting, chronically recurrent, and acutely explosive provide a novel construct for understanding bTB breakdown heterogeneity that moves beyond aggregate population-level descriptions. This distinction is well captured by characterising Cluster 2 as a “quiet herd” phenotype and Cluster 3 as the naïve, explosive phenotype: the former is quiet in any individual breakdown but chronically troubled, while the latter is dramatic and resource-intensive but not necessarily recurrent once clearance is achieved.

5.3. Cluster Stability and Methodological Robustness

The PAM $K = 3$ solution demonstrated exceptional stability across all validation approaches applied. Bootstrap resampling across $B = 200$ iterations produced mean Jaccard coefficients of 0.996, 0.998, and 0.982 for Clusters 1, 2, and 3 respectively, with zero dissolution events, substantially exceeding the 0.85 threshold for stable classification. Split-sample validation across five independent 70/30 partitions produced a mean ARI of 0.999 (SD = 0.002), confirming near-perfect recovery of the canonical solution on held-out data. Sequential variable removal identified spread ratio, breakdown duration, and prior breakdown history as the three primary structural drivers of the solution, with index test positivity and herd size contributing supplementary rather than defining information. Consensus clustering produced an ARI of 0.941 against the canonical PAM solution, with a clearly block-diagonal consensus matrix confirming that the three phenotypes are genuinely and reproducibly co-assigned across independent resamples. Together, these results satisfy the stability criteria required for an unsupervised classification intended for epidemiological inference and clinical application.

PAM was selected over k -means and hierarchical clustering on the grounds of its robustness to outlying observations and its identification of cluster representatives as actual observed breakdown events (medoids), making cluster prototypes directly interpretable in epidemiological terms. The Gaussian Mixture Model favoured a two-component solution, reflecting the geometric overlap between Cluster 1 and Cluster 2 in the feature space, but this coarser partition was considered insufficient to capture the mechanistically distinct recurrence phenotype. Fuzzy C-means confirmed that cluster boundaries are not perfectly discrete a finding that is biologically plausible given the spectrum of breakdown severity observed in national surveillance data but the high bootstrap stability and split-sample ARI confirm that the three-cluster partition captures genuine and reproducible structure despite this continuous variation.

5.4. Local Neighbourhood Infection Pressure as the Dominant Risk Factor

The multinomial regression analysis revealed a clear and consistent asymmetry between the two domains of epidemiological exposure examined. Neighbourhood infection pressure

particularly the number of neighbouring New Breakdown herds was the strongest and most consistent predictor of breakdown cluster membership across all three phenotypes, with each additional new breakdown neighbour associated with a 37–47% increase in the odds of belonging to a breakdown cluster relative to control herds. The gradient in odds ratios across clusters weakest for Cluster 1 and strongest for Cluster 3 suggests that neighbourhood infection pressure not only increases the probability of breakdown occurrence but preferentially drives membership in the more severe, acute high-intensity transmission phenotype. This is consistent with a local microepidemic model in which the surrounding epidemiological environment is a stronger determinant of breakdown phenotype than herd-level characteristics alone

In contrast, inward movement variables produced substantially smaller effect sizes, with per-unit odds ratios close to 1.00 for most predictors, indicating that the infection status of individual animal movements contributes less to breakdown phenotype than the local epidemiological context. This does not preclude a meaningful role for trade-mediated transmission at higher movement volumes, and future analyses modelling movement exposure as a categorical or polynomial term may reveal nonlinear dose-response relationships. The protective association of Control-status neighbouring herds across all three clusters may reflect genuine epidemiological buffering by an uninfected local environment, or alternatively may indicate that herds in low-prevalence areas are less likely to encounter the transmission pressure that drives the more severe breakdown phenotypes. Disentangling these explanations would require longitudinal data linking neighbourhood composition to subsequent breakdown incidence.

The significant interaction between neighbourhood infection pressure and calendar year confirms that these associations are not static the effect of new breakdown neighbours on cluster membership strengthened gradually over the study period, consistent with the fluctuating national bTB prevalence observed across the same timeframe. This temporal heterogeneity warrants further investigation in analyses with extended follow-up and explicit modelling of national prevalence trends as a time-varying covariate.

5.5. Survival and Recurrence Dynamics

Kaplan–Meier and Cox proportional hazards analyses confirmed that the three breakdown phenotypes differ not only in their within-breakdown characteristics but in their longitudinal trajectories. Cluster 2 herds experienced subsequent breakdowns at more than twice the rate of Cluster 1 herds after full adjustment ($HR = 2.370$), a finding that persisted after controlling for prior breakdown history confirming that the excess recurrence risk of this phenotype is not simply a function of its elevated prior history but reflects additional persistent risk

factors not captured by the covariates in the model. This is consistent with the hypothesis that Cluster 2 herds harbour residual infection or face continual low-level environmental or wildlife exposure that current control protocols do not adequately address. Prior breakdown history was by far the strongest predictor of recurrence ($HR = 21.131$), underscoring the importance of chronic herd-level susceptibility as a driver of the national bTB burden. The proportional hazards assumption was violated for both outcomes, reflecting the heterogeneous trajectories of the three phenotypes over extended follow-up, and results should be interpreted as average effects over the observation period.

5.6. Implications for Disease Control

These findings have direct implications for the design of targeted bTB control interventions. The identification of three mechanistically distinct breakdown phenotypes suggests that a uniform response to all bTB breakdowns is unlikely to be optimal. Cluster 1 herds self-limiting, rapidly resolving, and accounting for nearly two thirds of all breakdowns may be adequately managed under standard test-and-removal protocols without intensive additional intervention. Cluster 2 herds chronically recurrent despite limited within-herd spread represent the greatest challenge for eradication, as their recurrence pattern is not adequately explained by prior history alone and likely reflects persistent exposure sources that test-and-removal cannot address. These herds would benefit from intensified environmental investigation, enhanced wildlife management, and potentially more frequent surveillance intervals. Cluster 3 herds acutely explosive but not chronically recurrent require intensified short-term response to accelerate clearance and limit within-herd amplification, but may not require the same long-term sustained intervention as Cluster 2. The dominance of neighbourhood infection pressure as a predictor of cluster membership additionally supports spatial targeting of control resources toward areas of local microepidemic activity, where the density of actively breaking-down herds creates transmission pressure that elevates breakdown risk and severity for all herds in the surrounding landscape.

5.7. Limitations

Several limitations of this study warrant acknowledgement. The proportional hazards assumption was violated in both Cox models, indicating that hazard ratios varied over the follow-up period and should be interpreted as average rather than constant effects. The multinomial model achieved moderate discrimination (Hand–Till multiclass AUC = 0.626), indicating that movement and neighbourhood variables capture meaningful but incomplete information about breakdown phenotype unmeasured factors such as farm biosecurity, wildlife interface intensity, and within-herd immune status likely contribute to the residual variation. Controlled finishing unit herds and breakdowns of fewer than 60 days were excluded from the

analytical cohort; sensitivity analyses including these groups, as suggested by collaborators, would clarify whether the identified phenotypes generalise to the full breakdown population. The year interaction model was restricted to breakdown herds, precluding direct assessment of whether neighbourhood effects on breakdown occurrence also varied over time. Finally, the spatial radius used to define neighbourhood herds was fixed; varying this threshold may alter the estimated neighbourhood effects and warrants exploration in future analyses.

5.8. Conclusions

This study demonstrates that bTB breakdowns in Irish cattle herds are epidemiologically heterogeneous, with three distinct phenotypes reflecting fundamentally different underlying mechanisms of infection, persistence, and reinfection. The identified phenotypes are stable, reproducible, and independently validated across multiple analytical approaches. Neighbourhood infection pressure is the dominant determinant of breakdown phenotype, pointing to a local microepidemic phenomenon in which the surrounding epidemiological context shapes not only whether a breakdown occurs but what form it takes. Phenotype-informed intervention with intensified environmental investigation for chronically recurrent herds and rapid response for acutely explosive phenotypes represents a more targeted and resource-efficient approach to bTB control than uniform response protocols, and provides a framework for directing the substantial annual investment in Ireland’s eradication programme toward the herds and breakdown types where intervention is most likely to have impact.

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Figures

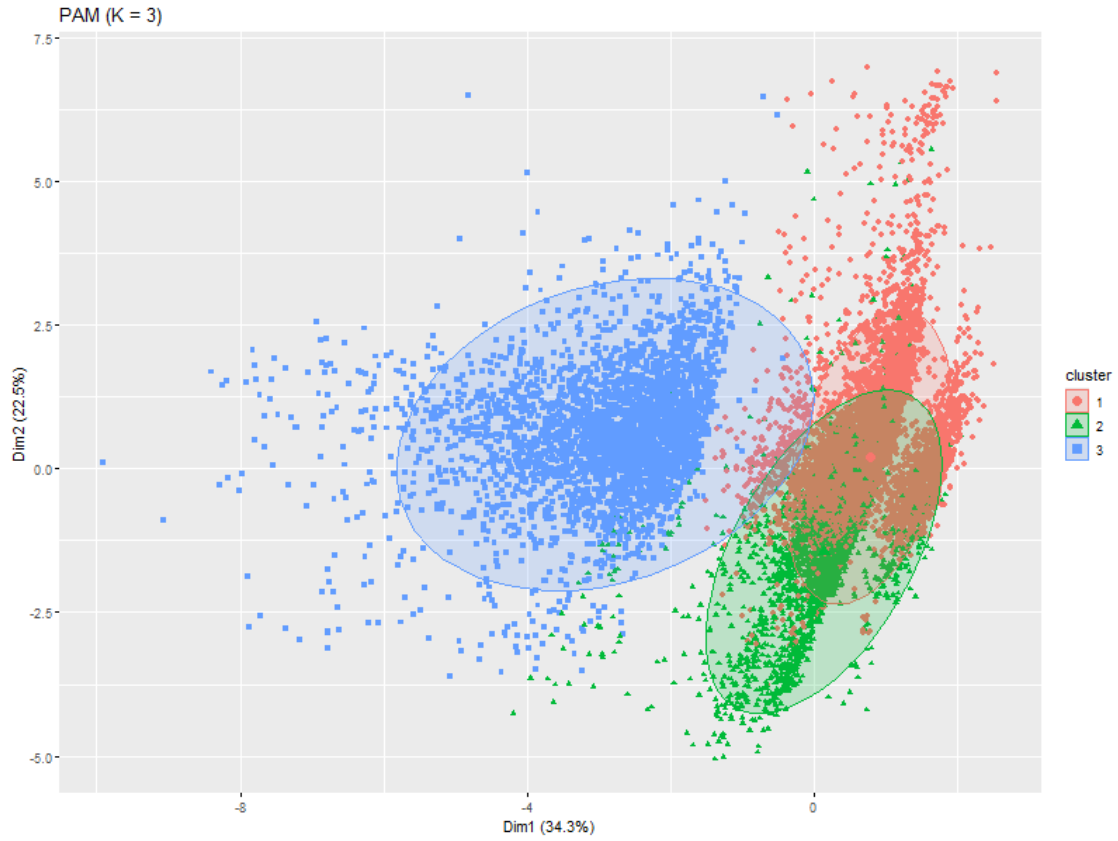


Figure 4: PAM clustering results for $K = 3$ projected onto the first two principal components. Clusters are coloured and surrounded by 95% confidence ellipses. The three clusters exhibit moderate separation in the two-dimensional PCA projection, with Cluster 3 occupying the positive direction of PC1, consistent with its elevated spread ratio and prolonged breakdown duration.

Supplementary Materials 1

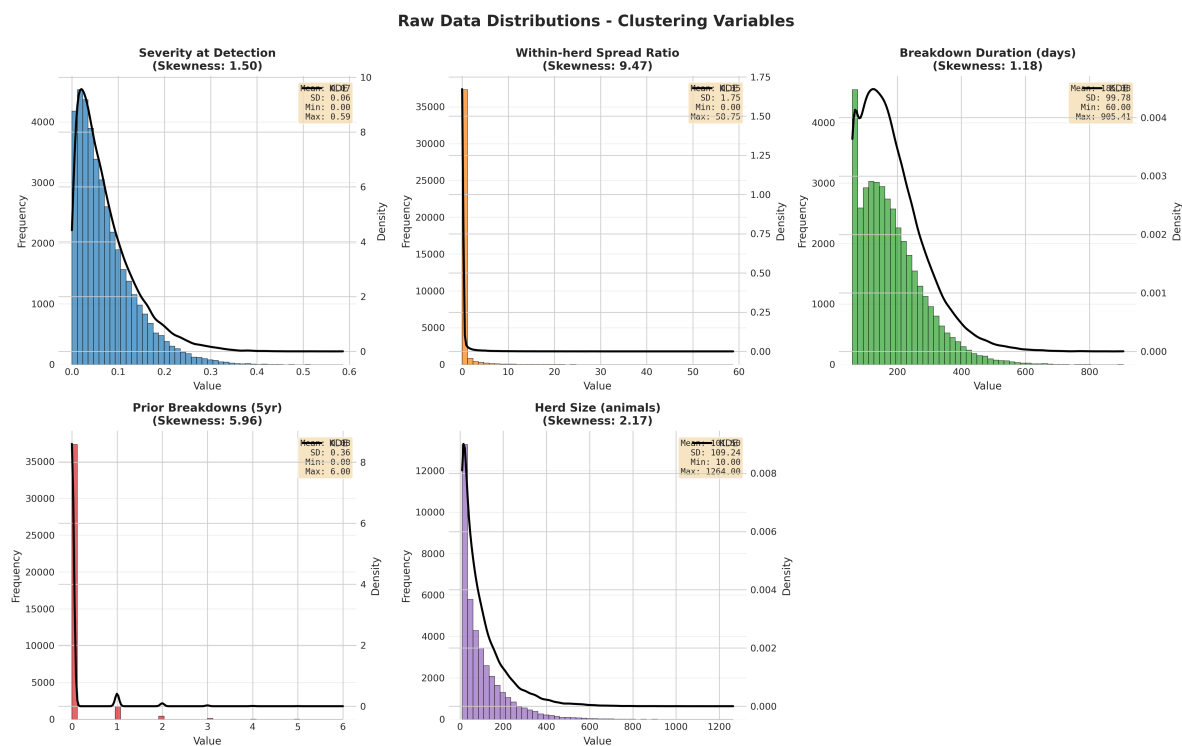


Figure 5: Histograms of raw variables prior to transformation, showing right-skewness and zero inflation in several variables.

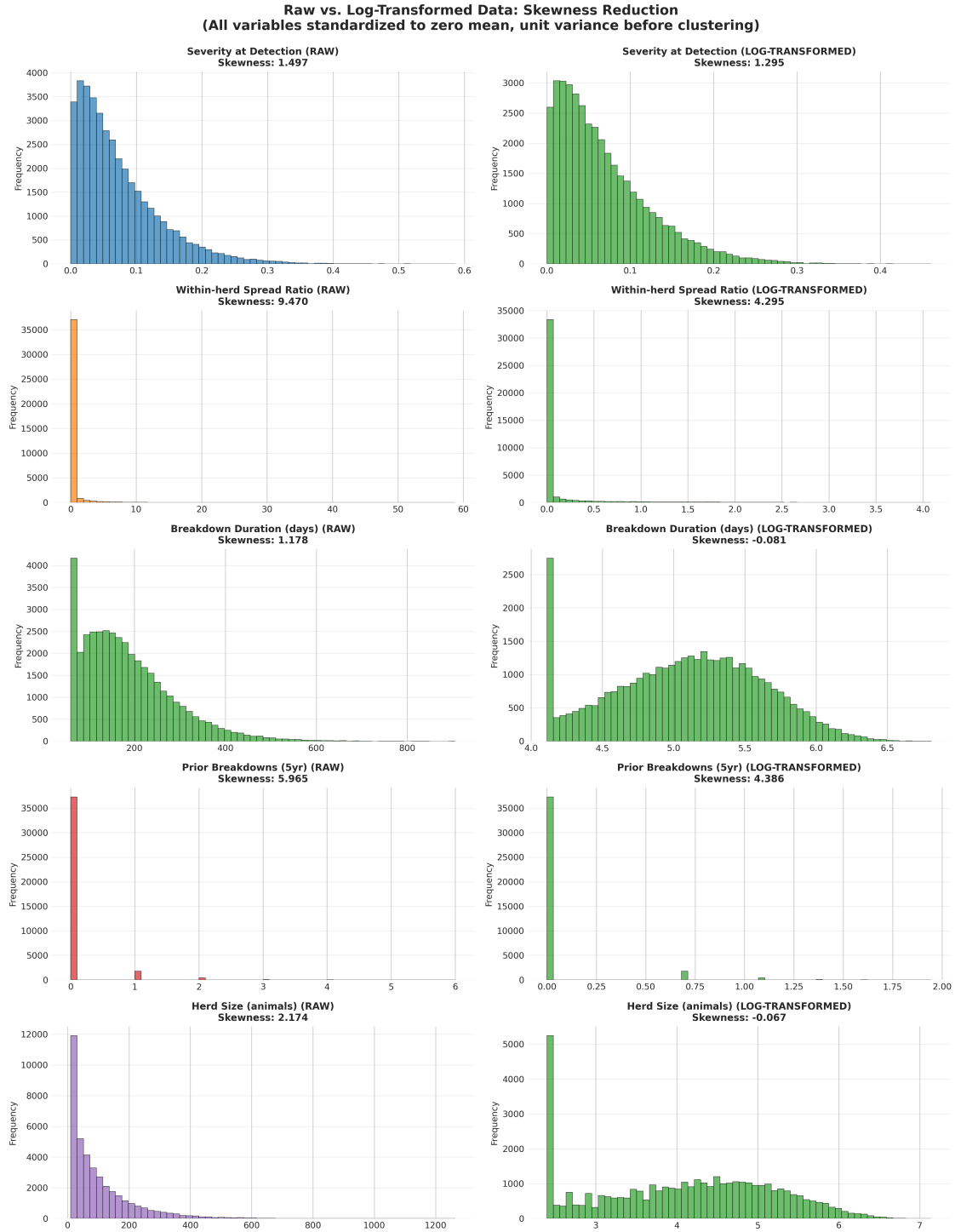


Figure 6: Histograms of variables before vs after log-transformation.

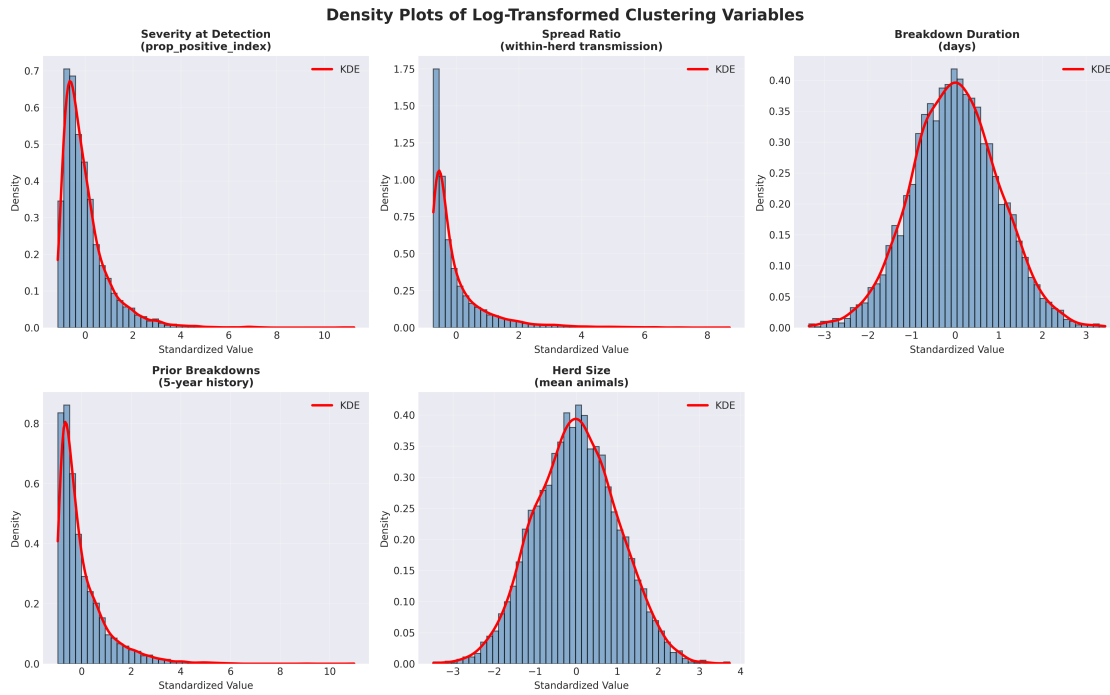


Figure 7: Density plots of transformed variables.

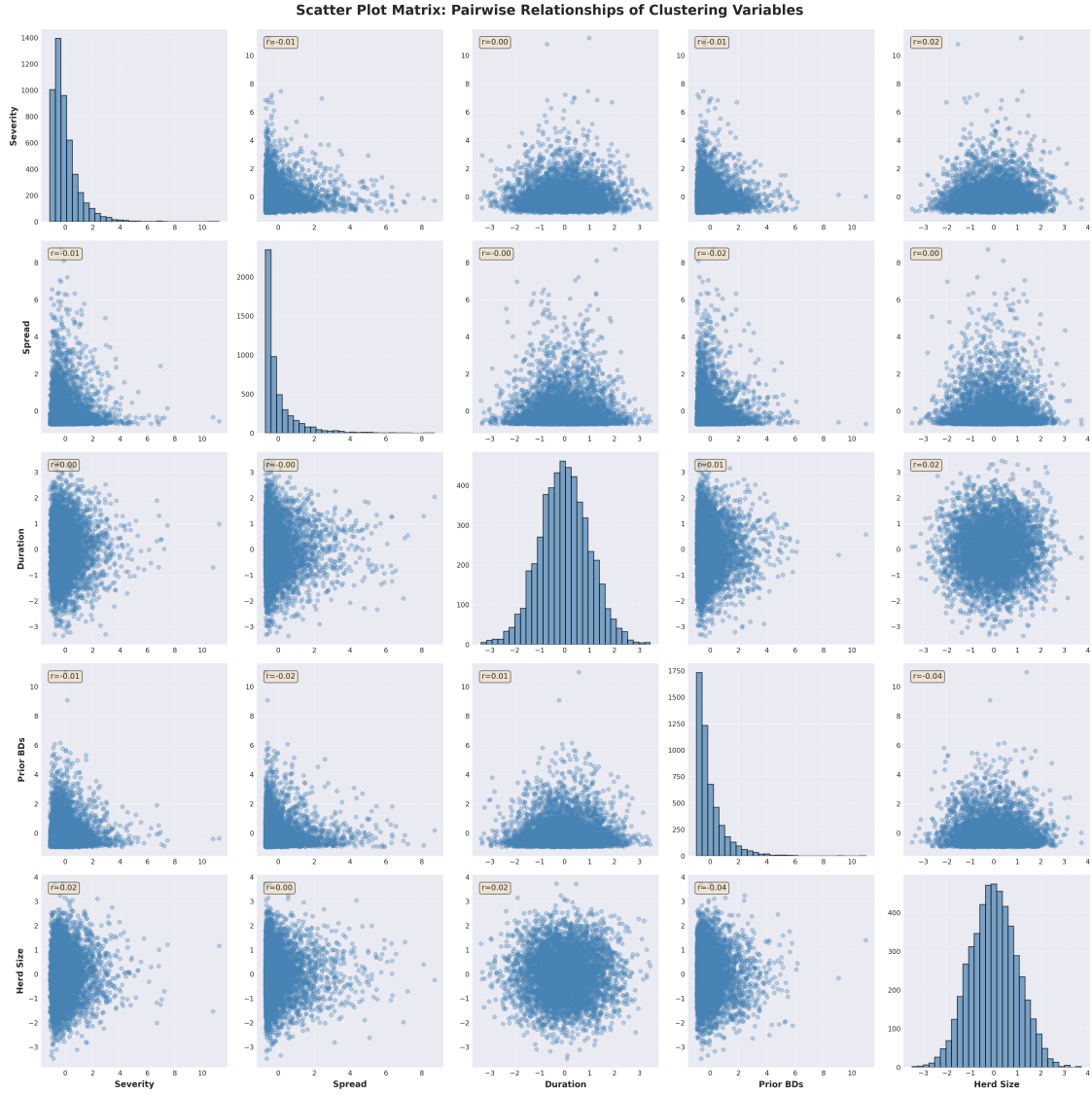


Figure 8: Scatter plot matrix illustrating relationships between variables.

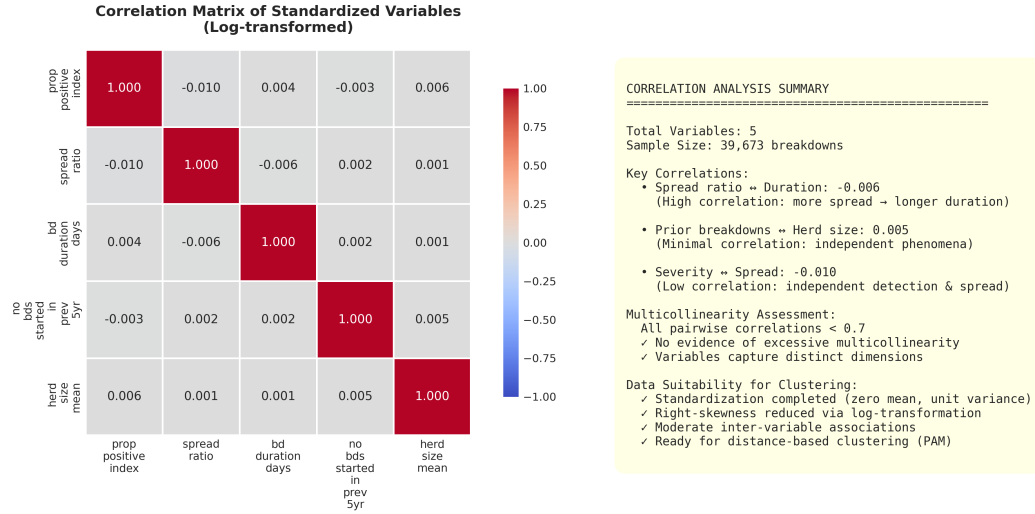


Figure 9: Correlation matrix of variables used in the clustering analysis.

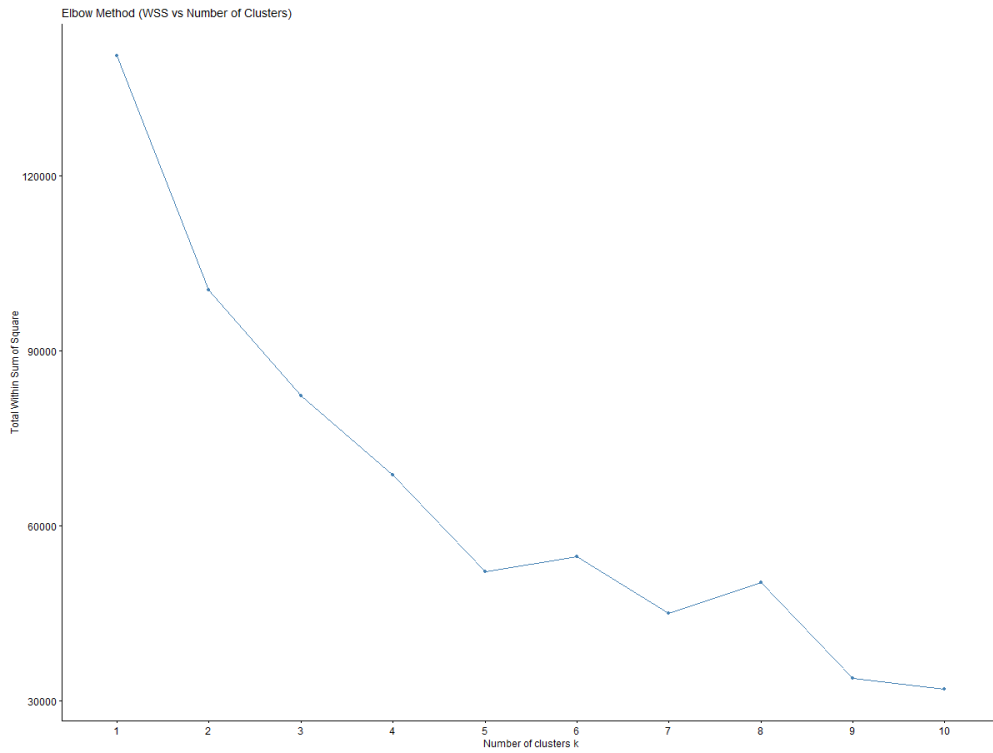


Figure 10: Elbow method based on within-cluster sum of squares (WSS) for $K = 1$ to 10. The curve shows a steep decline up to $K = 3$, followed by a plateau, indicating diminishing improvements in cluster compactness beyond this point.

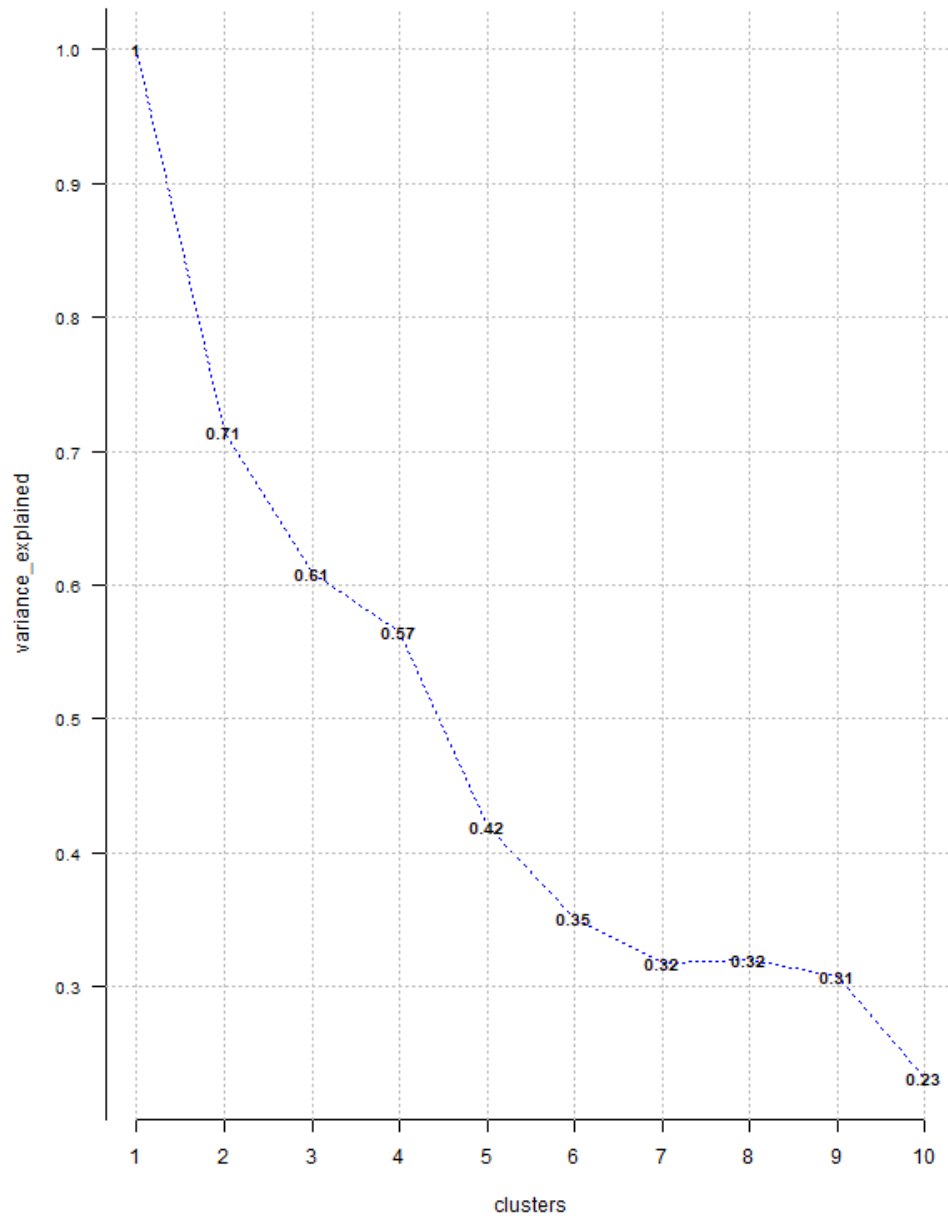


Figure 11: Normalised cluster compactness from the `ClusterR` elbow method.

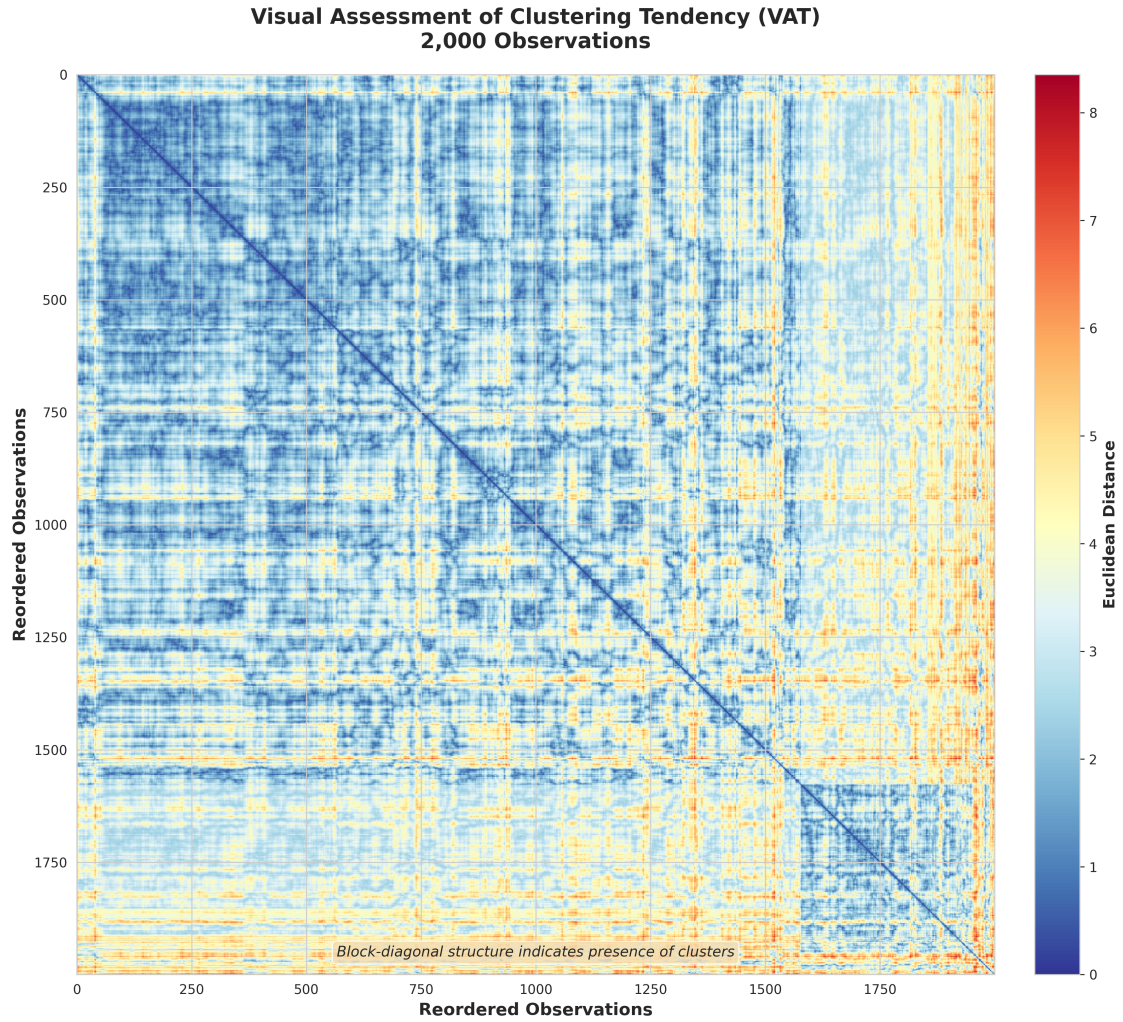


Figure 12: Visual Assessment of Cluster Tendency (VAT) plot. The reordered dissimilarity matrix exhibits distinct block-diagonal structures, indicating the presence of well-separated clusters within the data.

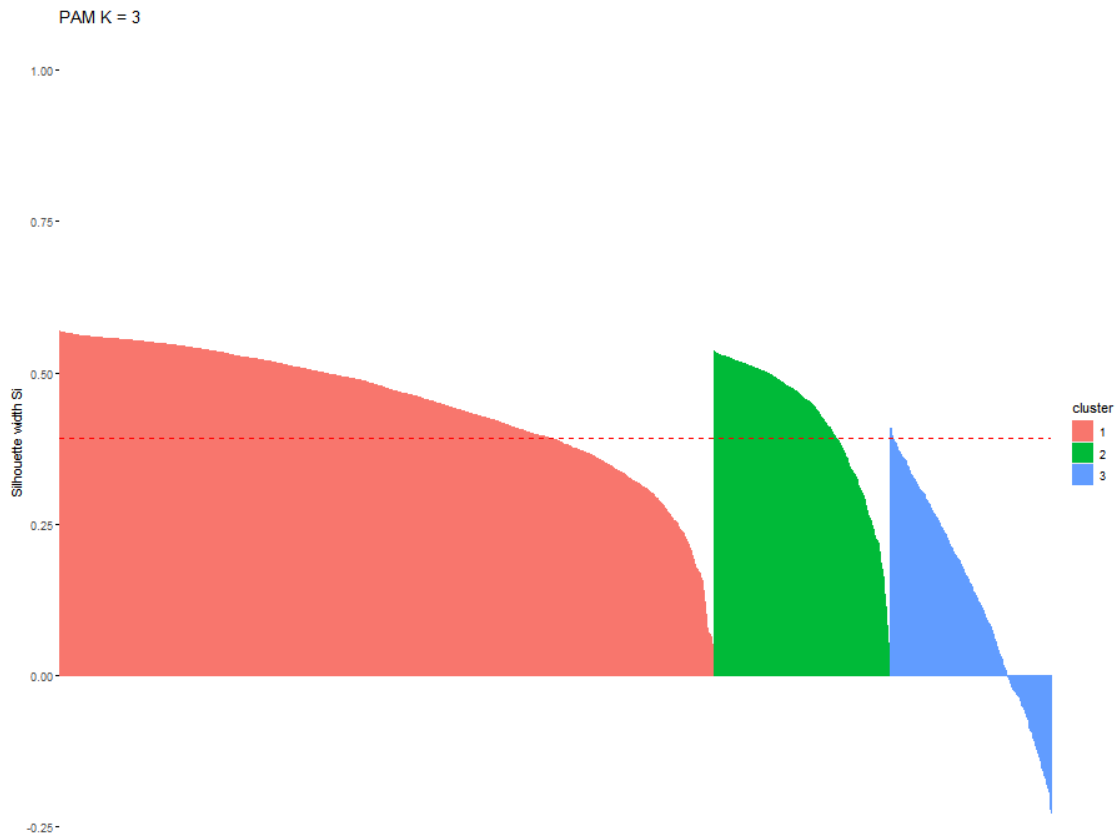


Figure 13: Silhouette plots for PAM clustering with $K = 3$. The dashed line indicates the average silhouette width.

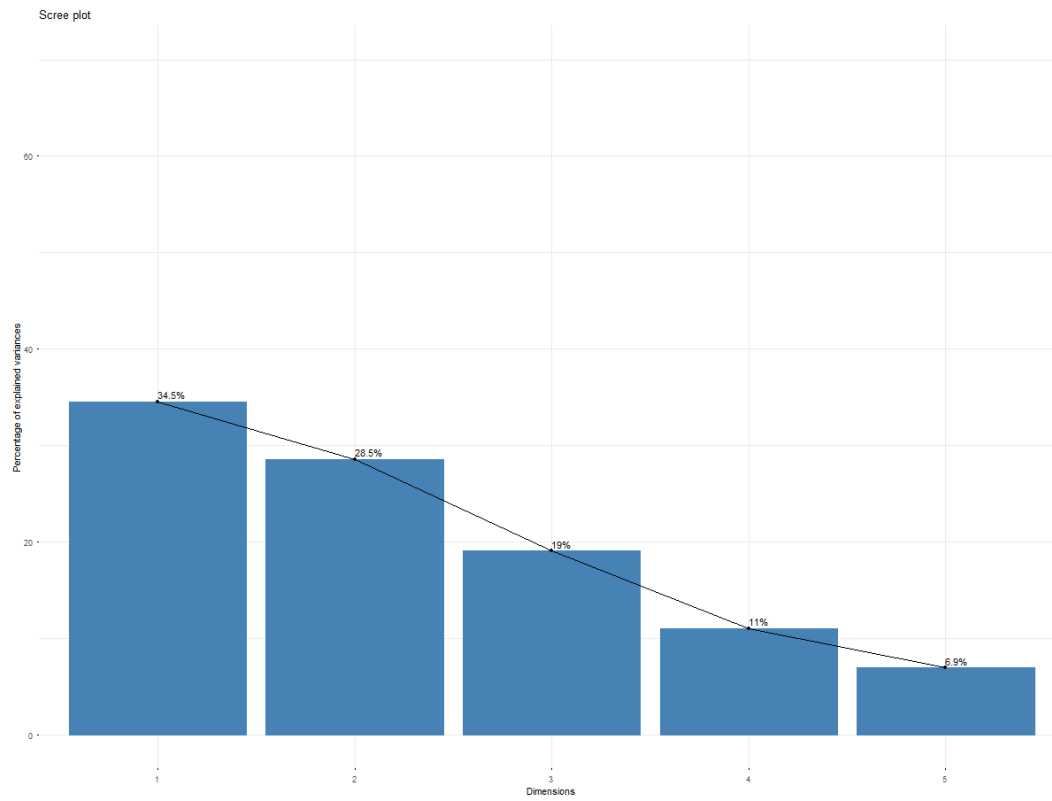


Figure 14: Scree plot illustrating the percentage of total variance explained by each principal component. Values above bars indicate the percentage of variance explained by each component individually.

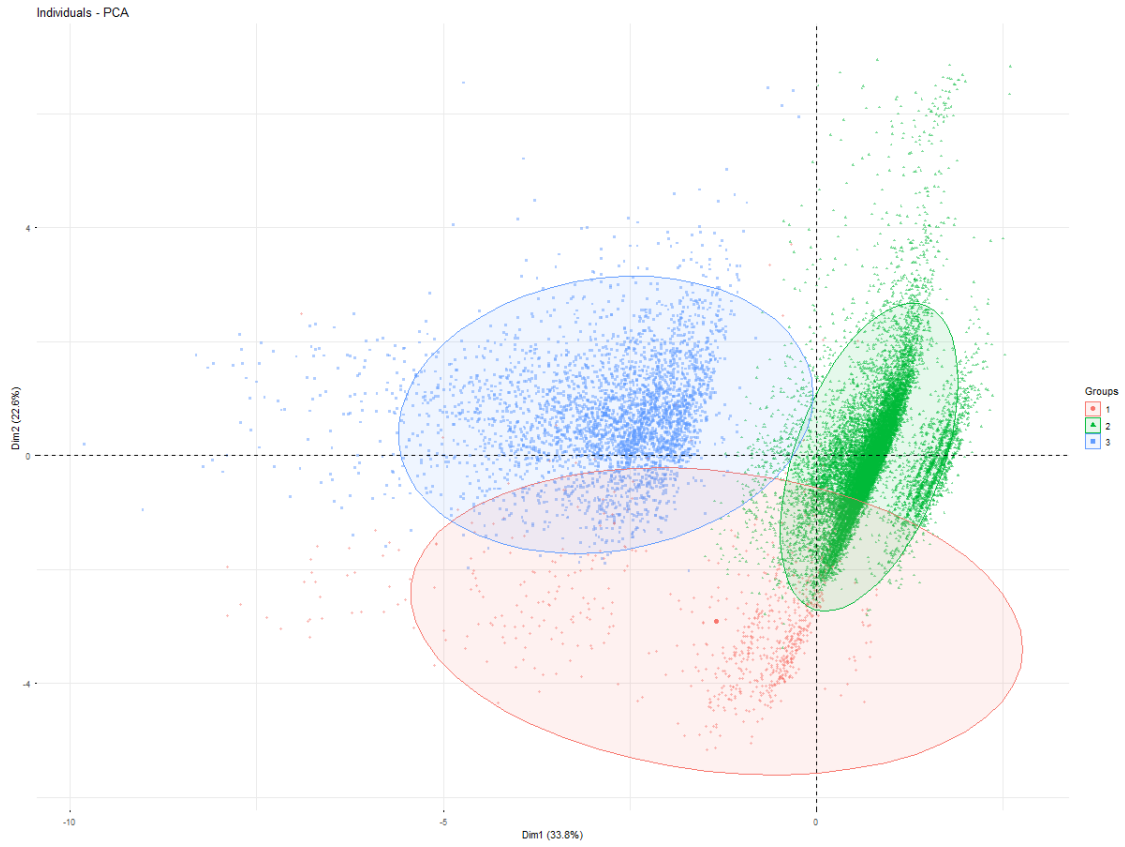


Figure 15: Clustering results for $K = 3$ projected onto the first two principal components.

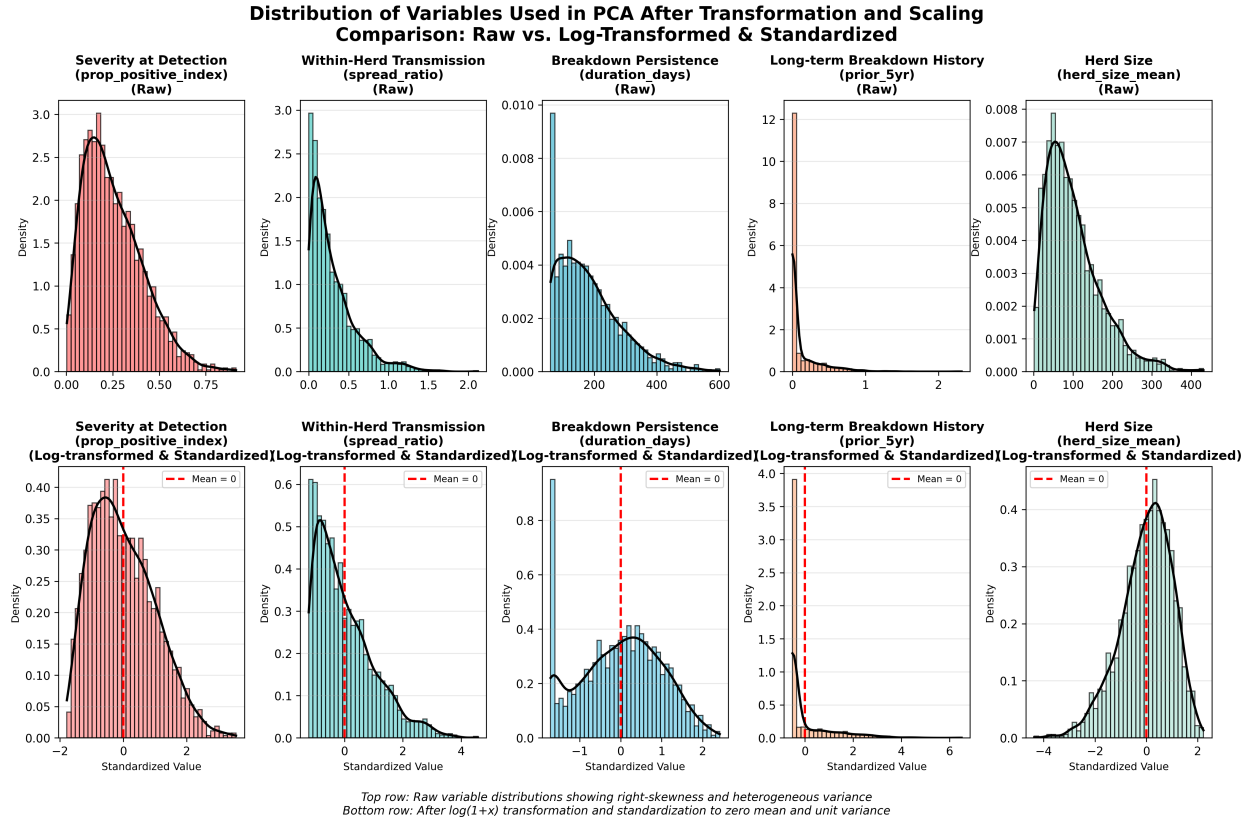


Figure 16: Distribution of variables used in PCA before vs after transformation and scaling.

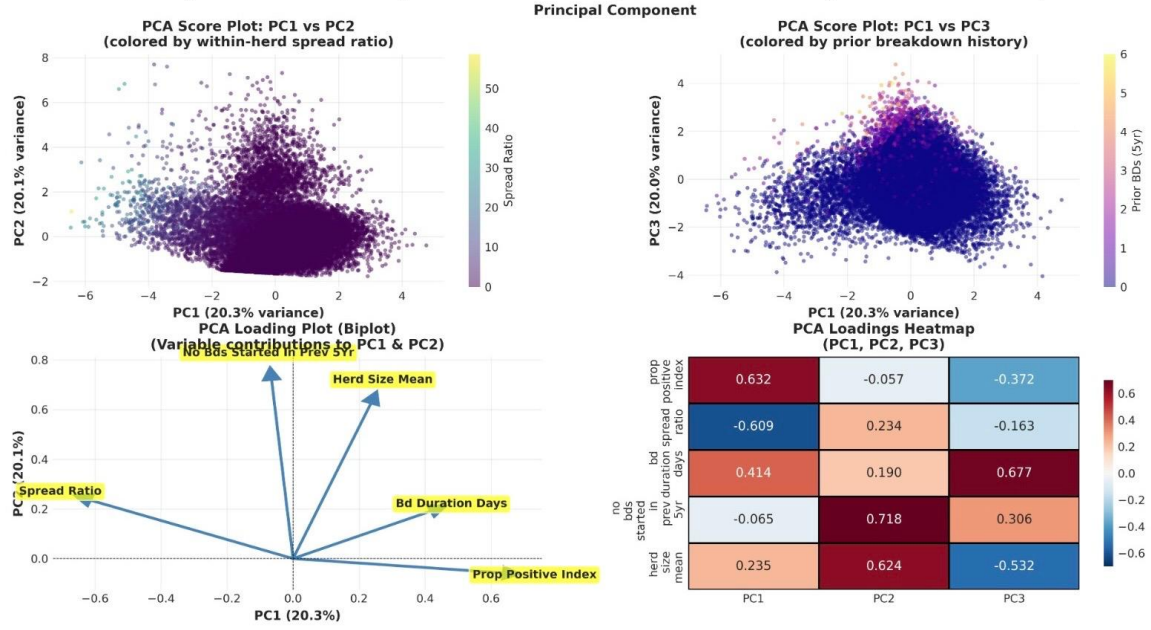


Figure 17: PCA analysis of the standardised clustering feature matrix. The figure presents the distribution of observations projected onto the first two principal components (PC1 and PC2), the PCA biplot illustrating variable loadings, and cluster assignments from PAM ($K = 3$). The point cloud exhibits a concentrated central region with increasing dispersion along PC1, reflecting a dominant axis of variation corresponding to acute transmission severity and breakdown persistence. Variables associated with transmission intensity breakdown duration and spread ratio are aligned along the negative direction of PC1, while index test positivity and herd size contribute more strongly to PC2. Cluster assignments reveal moderate separation in the two-dimensional projection, with one cluster concentrated along the positive direction of PC1 and the remaining clusters occupying the central and negative regions, with some overlap consistent with the continuous rather than discrete nature of bTB breakdown dynamics.

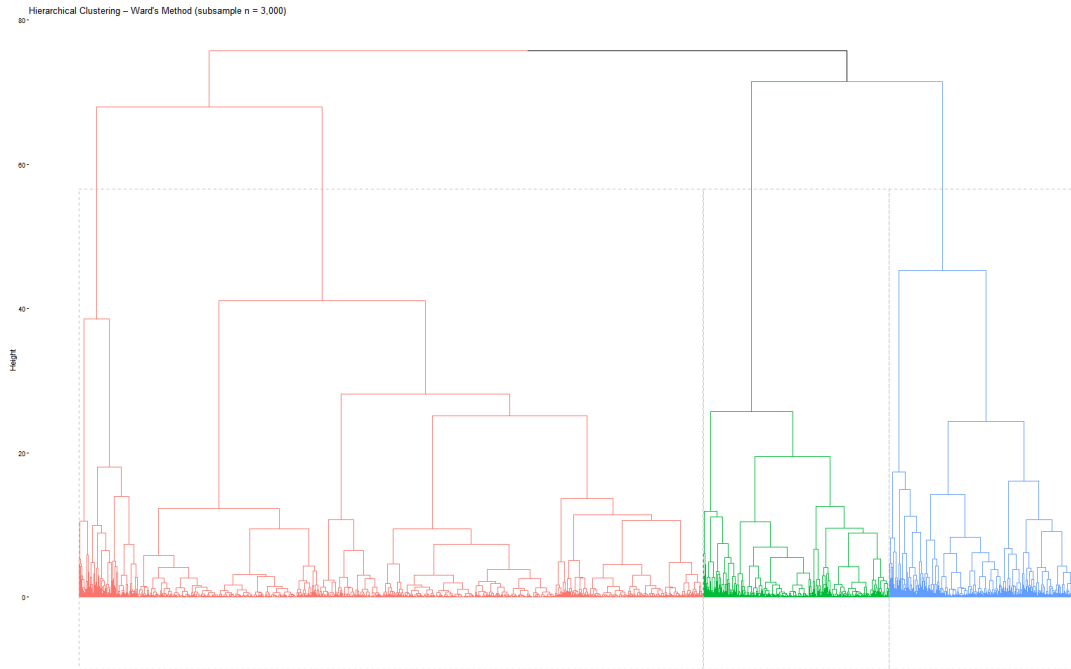


Figure 18: Hierarchical clustering dendrogram illustrating the relationships among observations. The dendrogram was examined to assess the consistency of the hierarchical structure with the three-cluster solution obtained using the PAM algorithm.

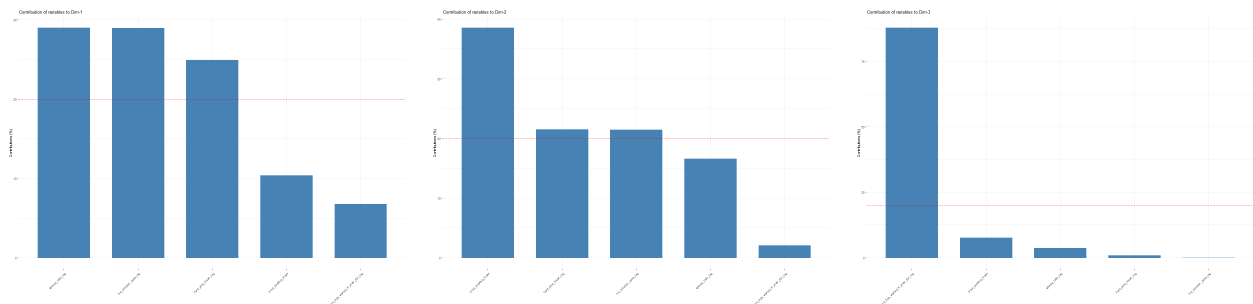


Figure 19: Contribution of variables to the first three principal components (PC1, PC2, and PC3).

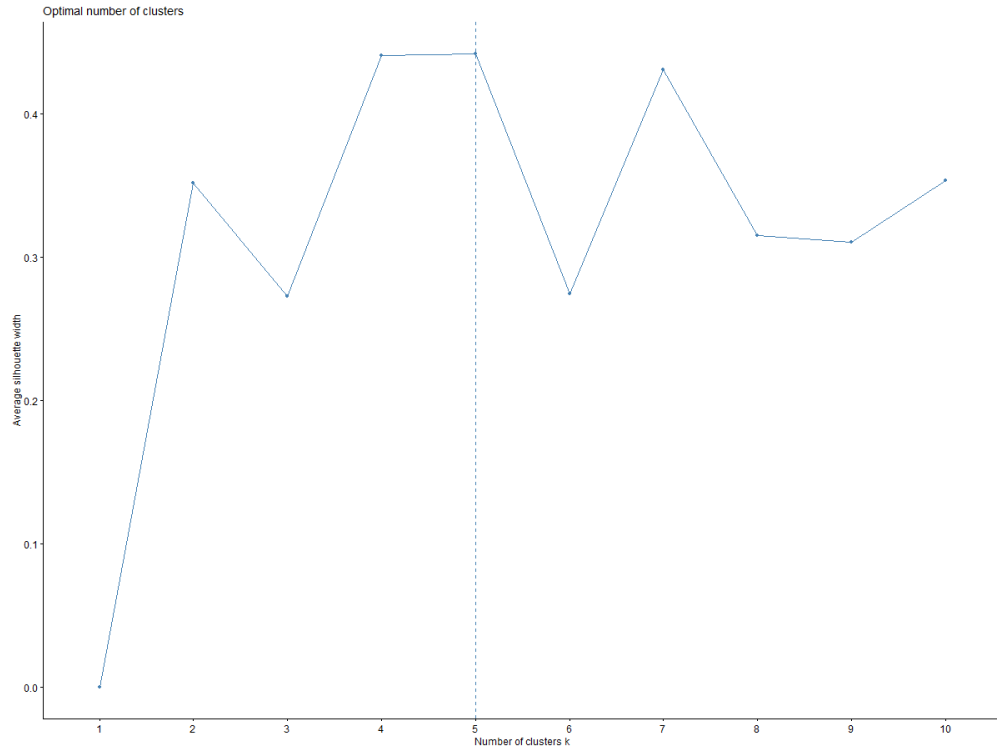


Figure 20: Average silhouette width for $K = 2$ to $K = 10$ based on k-means clustering.

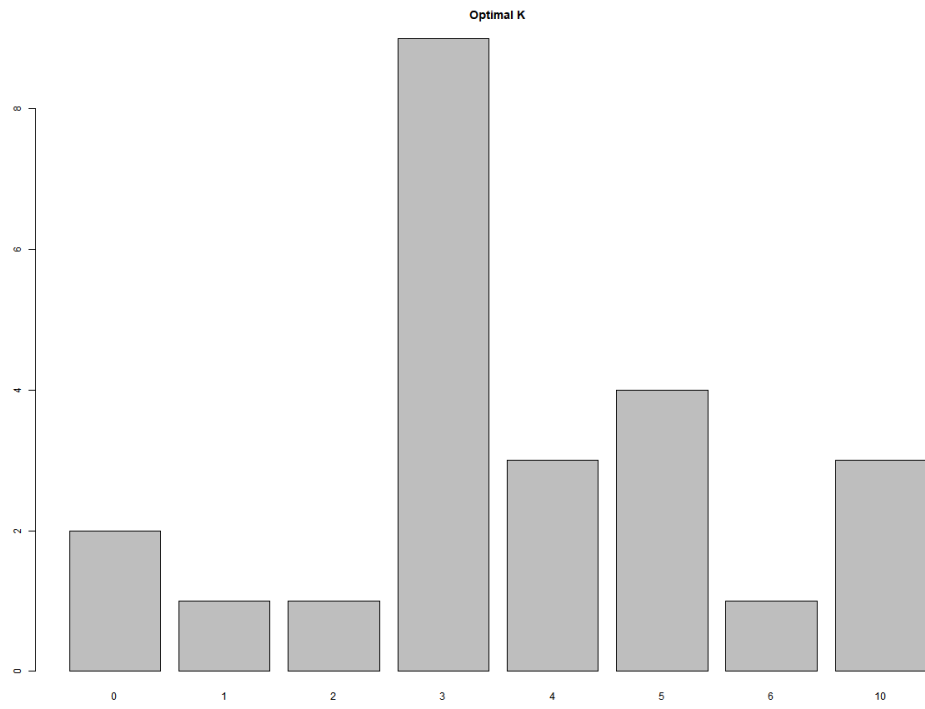


Figure 21: Number of clustering indices supporting each value of K from the NbClust analysis.

The results indicate that $K = 3$ is the most frequently recommended solution across the evaluated indices, receiving the highest level of support. While alternative values such as $K = 4$ and $K = 5$ are suggested by a smaller number of indices, their support is notably weaker.

This consensus across multiple independent criteria provides strong evidence that a three-cluster solution best captures the underlying structure of the data.

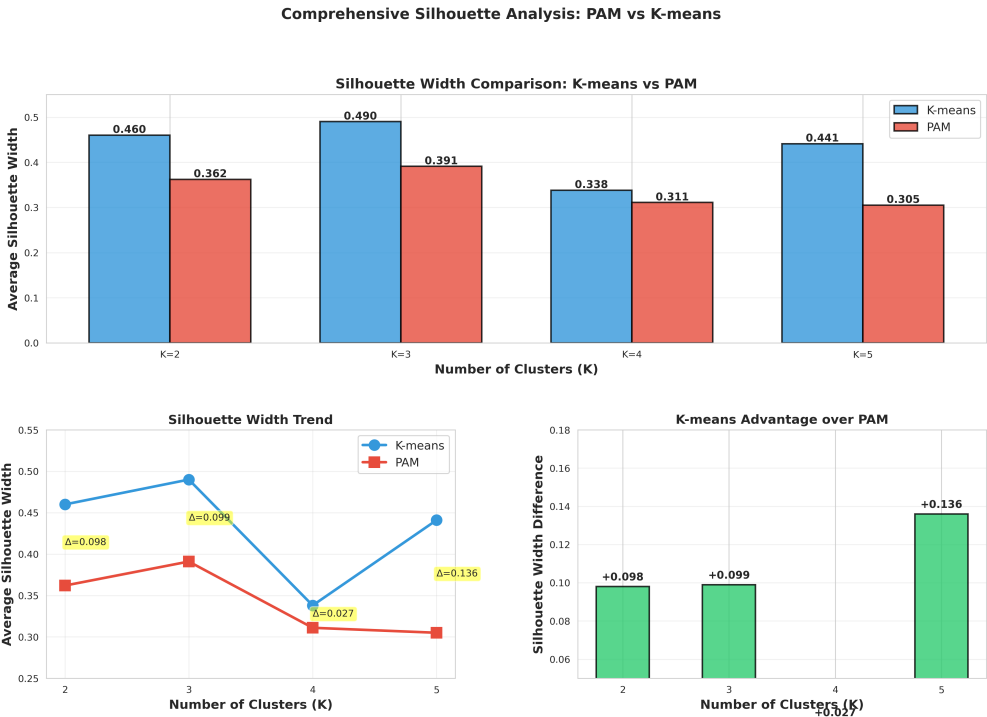


Figure 22: Average silhouette width for K -means and PAM across $K = 2, 3$, and 4 , computed on a random subsample of 5,000 observations from the standardised feature matrix.

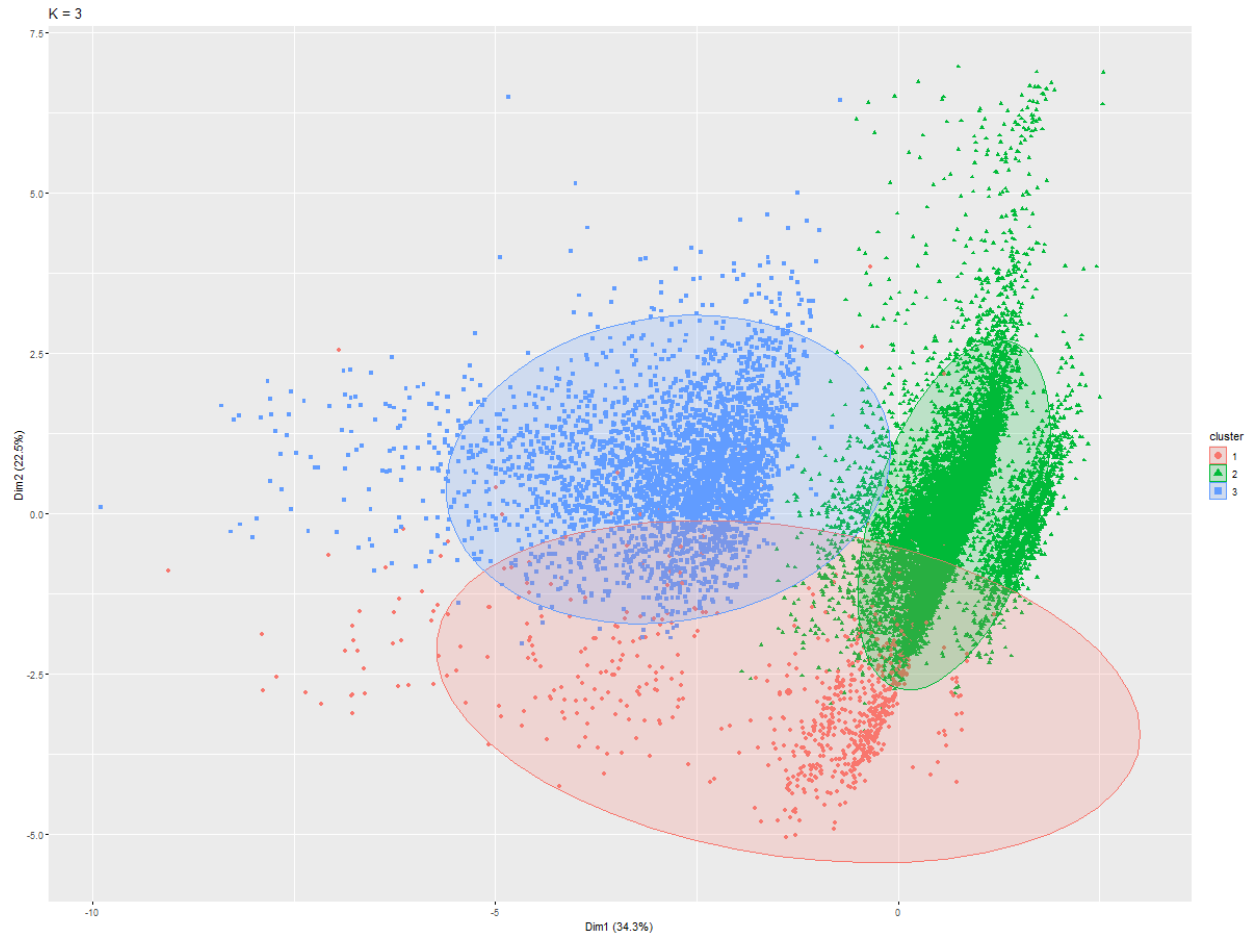


Figure 23: K-means clustering results for $K = 3$. Clusters are visualised using elliptical contours for clarity.

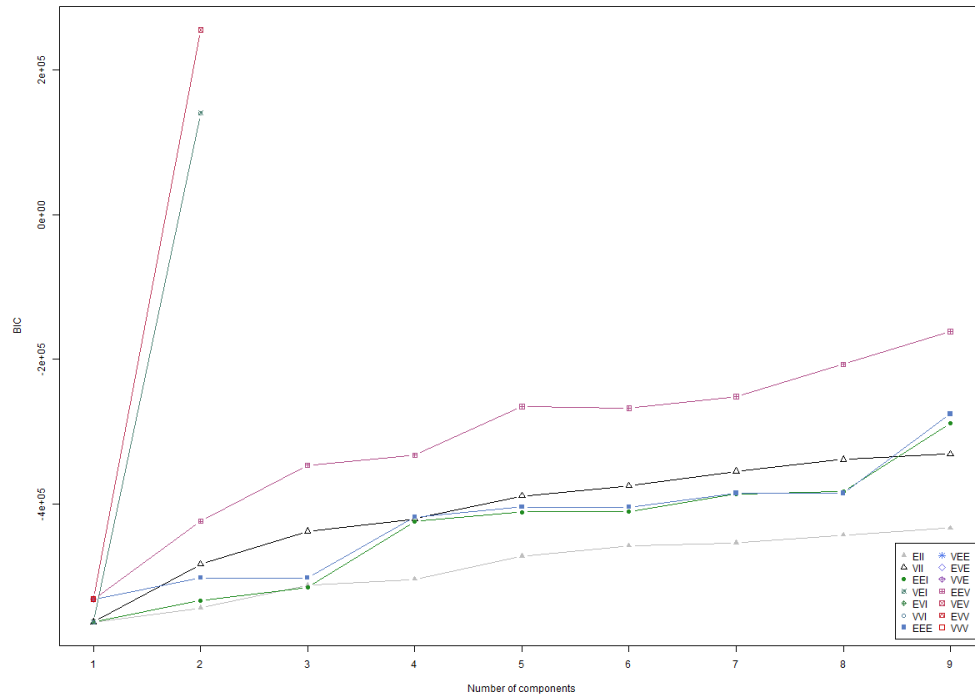


Figure 24: BIC values for Gaussian Mixture Models fitted across varying numbers of components and covariance structures. The optimal solution, indicated by the highest BIC, corresponds to a two-component model with ellipsoidal equal shape covariance structure (VEV).

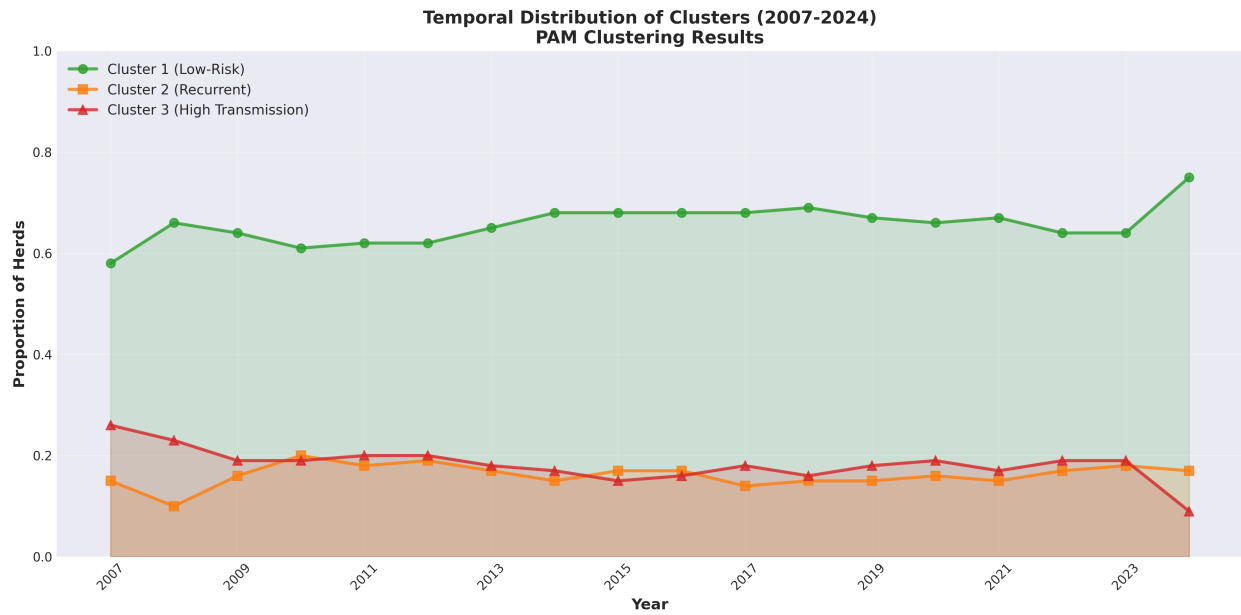


Figure 25: Temporal distribution of PAM clusters from 2007 to 2024, showing the proportion of herds assigned to each cluster over time.

Figure 25 illustrates the temporal distribution of clusters across the study period. Cluster

1 consistently represents the majority of herds, indicating that low-risk breakdown profiles dominate over time. Clusters 2 and 3 remain relatively stable but at lower proportions, with modest fluctuations across years. Notably, Cluster 3 shows a decline in the most recent year, while Cluster 1 increases, suggesting a potential shift toward less severe breakdown dynamics. Overall, the cluster structure remains broadly stable over time, supporting the robustness of the identified typologies.

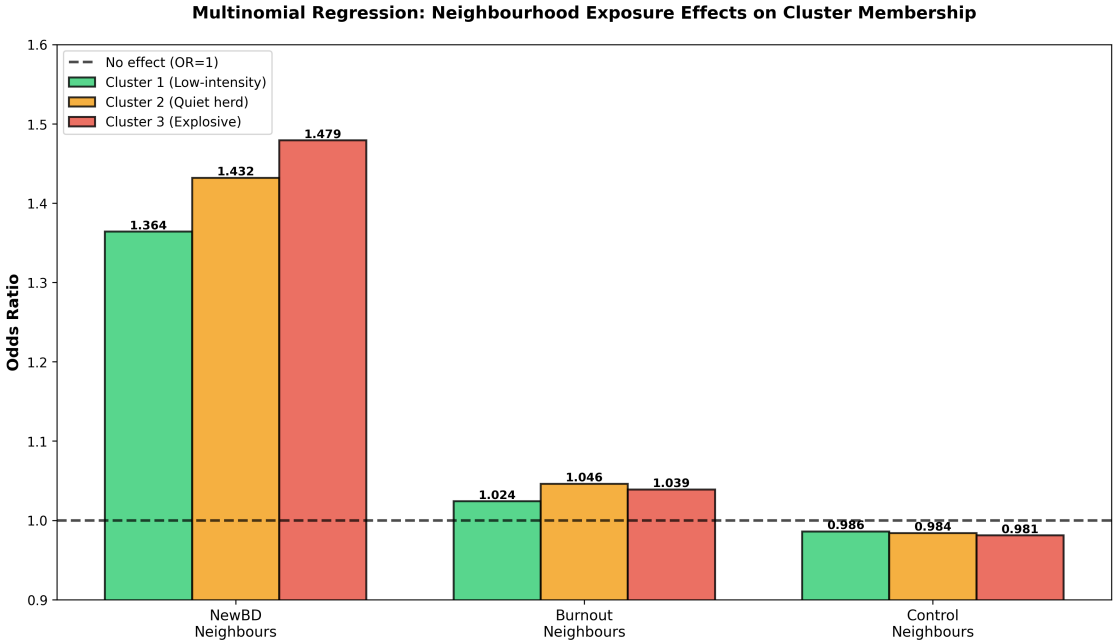


Figure 26: Odds ratios and 95% confidence intervals for neighbourhood infection pressure predictors from the multinomial logistic regression model. Estimates are presented for each breakdown cluster (Clusters 1–3) relative to control herds (Cluster 0). Values greater than 1 indicate increased odds of belonging to the corresponding breakdown phenotype relative to remaining breakdown-free.

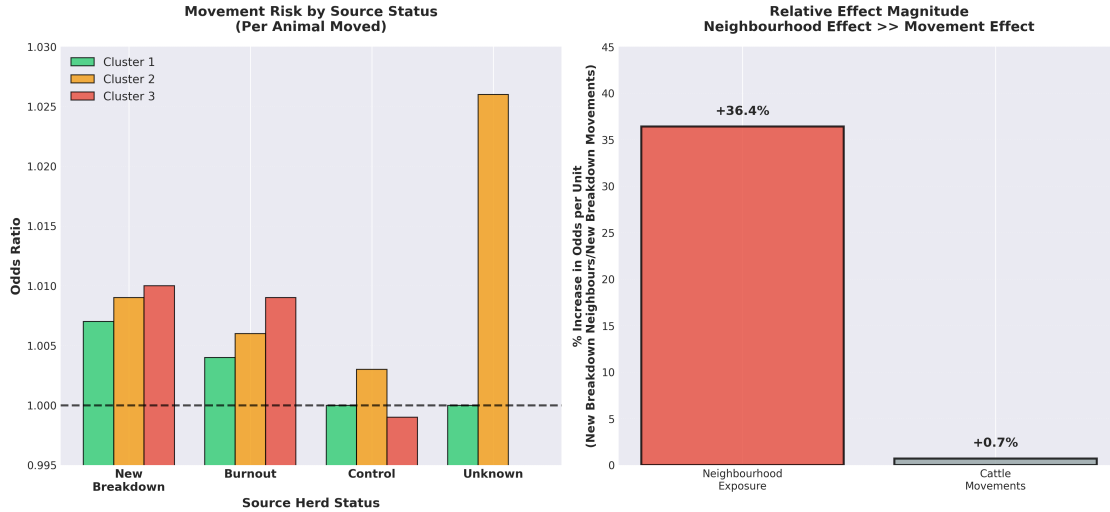


Figure 27: Odds ratios and 95% confidence intervals for inward movement predictors from the multinomial logistic regression model. Estimates are presented for each breakdown cluster (Clusters 1–3) relative to control herds (Cluster 0), reflecting the association between the bTB status of source herds for inward cattle movements and the probability of breakdown cluster membership.

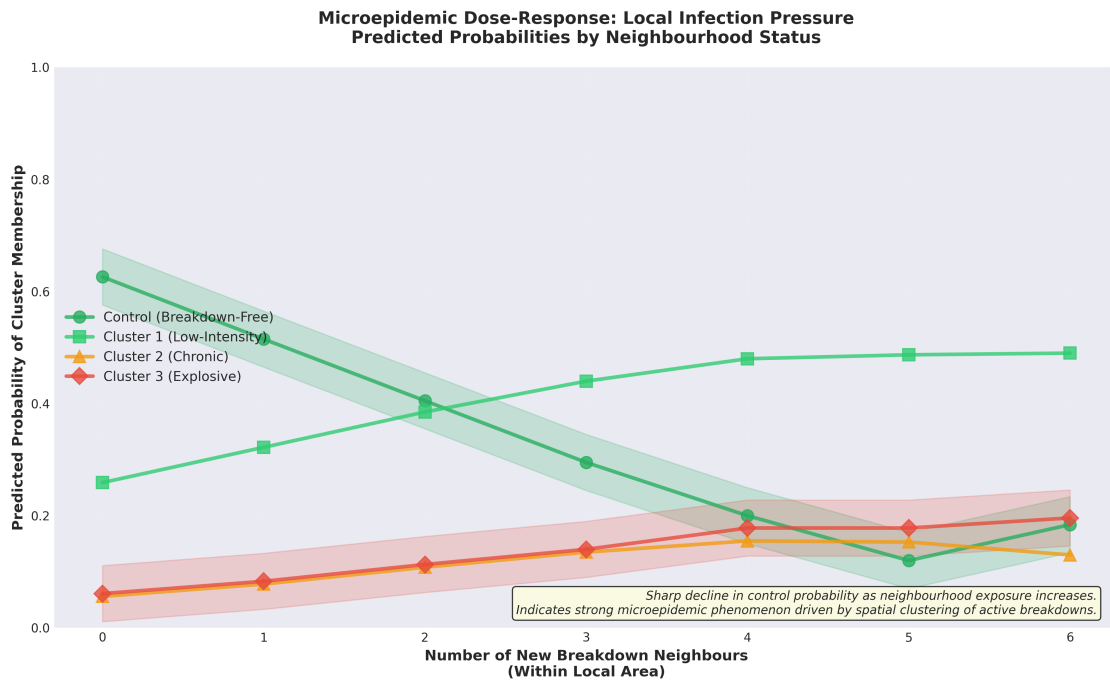


Figure 28: Predicted probabilities of cluster membership at the 10th, 50th, and 90th percentiles of each neighbourhood infection pressure predictor, with all other predictors held at their observed means. Probabilities are shown for control herds (Cluster 0) and each of the three breakdown phenotypes (Clusters 1–3), illustrating the marginal effect of neighbourhood infection pressure on the probability of breakdown occurrence and phenotype.

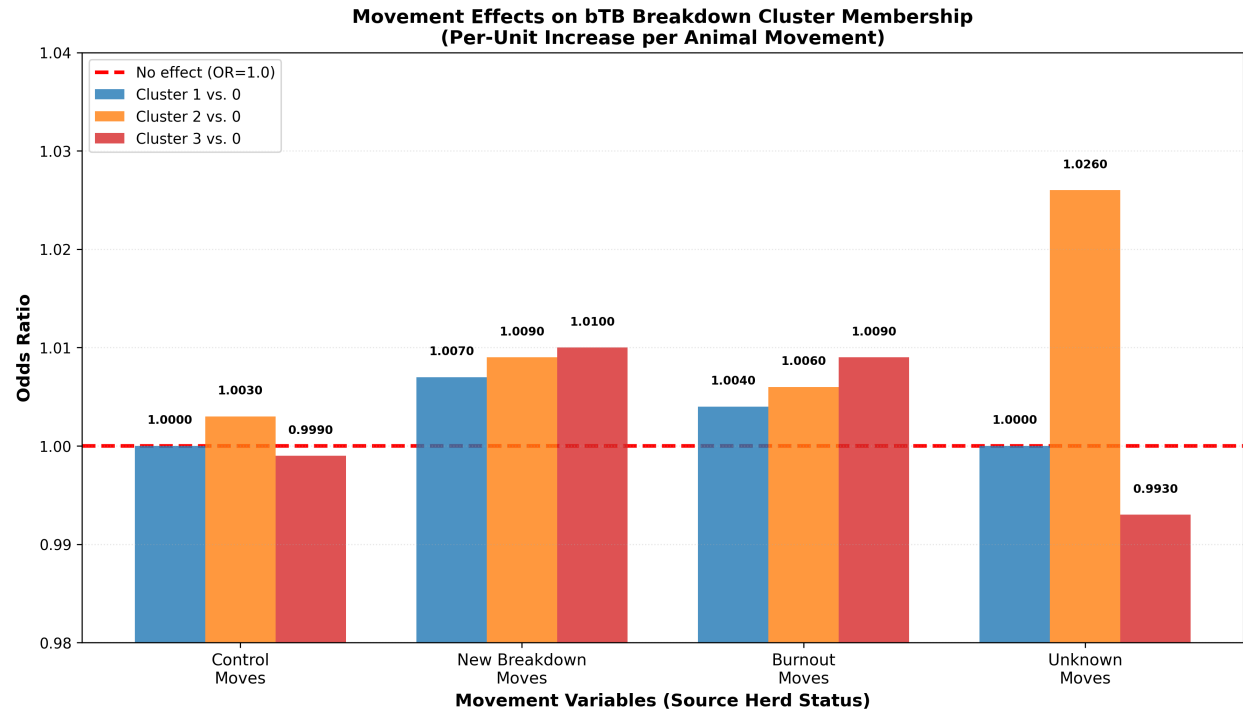


Figure 29: Movement effects on bTB breakdown cluster membership.

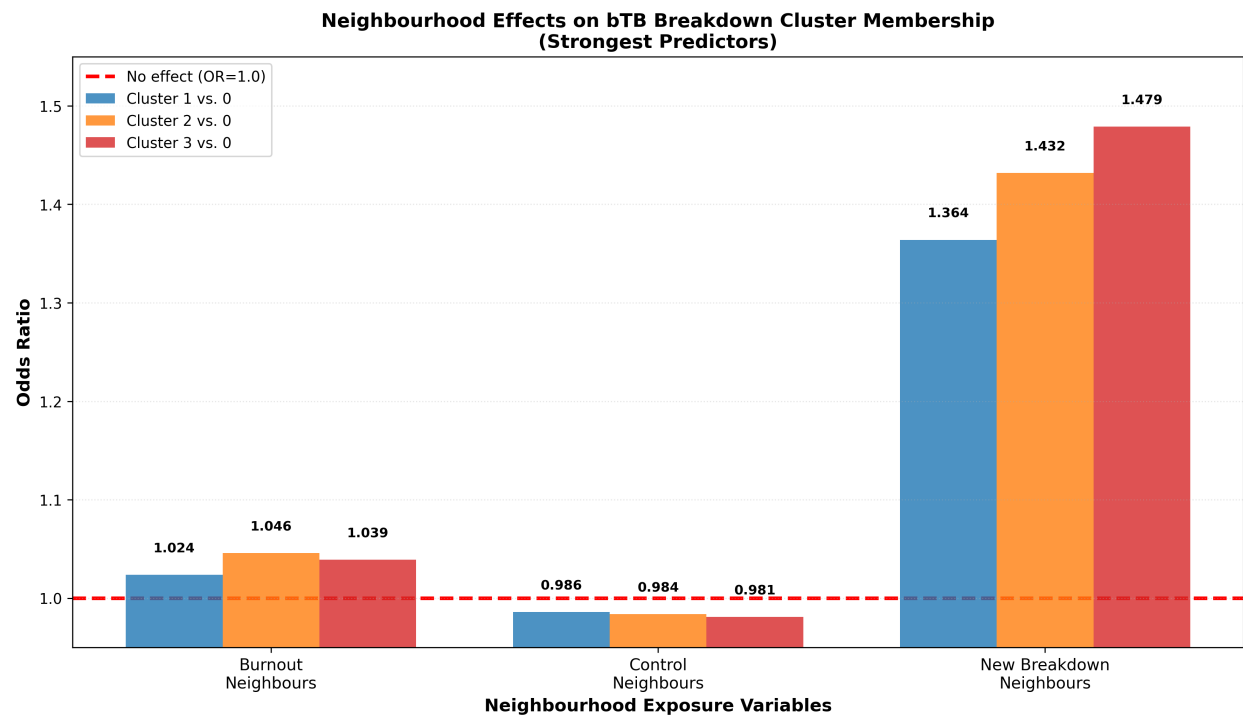


Figure 30: Neighbourhood effects on bTB breakdown cluster membership.

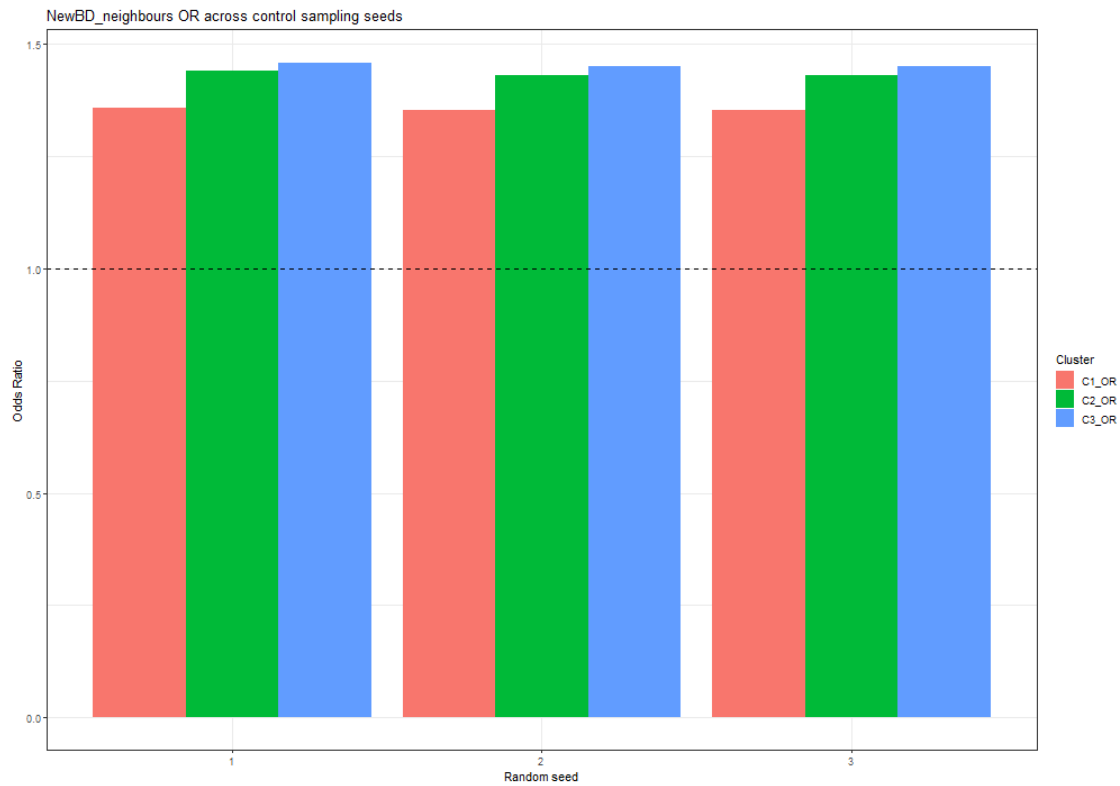


Figure 31: Odds ratios for **NewBD_neighbours** across three independent random seeds used for control herd sampling. The near-identical bars across seeds confirm that findings are not sensitive to the particular control sample drawn.

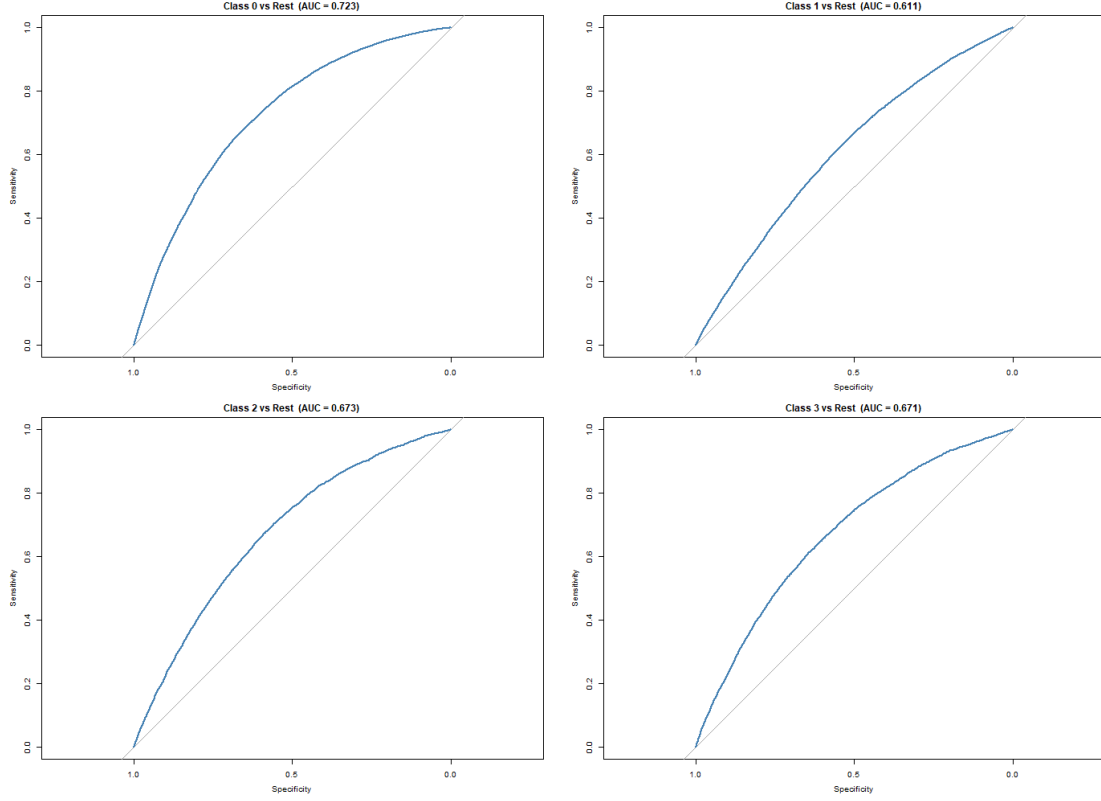


Figure 32: One-versus-rest receiver operating characteristic (ROC) curves for each outcome class from the multinomial logistic regression model. AUC values: Cluster 0 (control) = 0.723; Cluster 1 = 0.611; Cluster 2 = 0.673; Cluster 3 = 0.671. Hand-Till multiclass AUC = 0.626.

Table 8: Schoenfeld residuals test for the proportional hazards assumption in the breakdown duration Cox model.

Variable	χ^2	df	p
Cluster	1973.7	2	<0.0001
log(herd size)	133.7	1	<0.0001
log(prior BDs)	375.2	1	<0.0001
Short-term recurrence	12.1	1	0.0005
GLOBAL	2116.4	5	<0.0001

Table 9: Cox proportional hazards model for time to subsequent breakdown. Hazard ratios (HR) greater than 1 indicate faster recurrence relative to Cluster 1. Model fitted to $n = 39,673$ observations with $n = 18,675$ events. Concordance = 0.842.

Variable	HR	95% CI	z	p
Cluster 2	2.370	(2.177, 2.580)	19.89	<0.0001
Cluster 3	1.339	(1.282, 1.399)	13.16	<0.0001
log(herd size)	1.051	(1.036, 1.067)	6.60	<0.0001
log(prior BDs)	21.131	(18.959, 23.552)	55.12	<0.0001

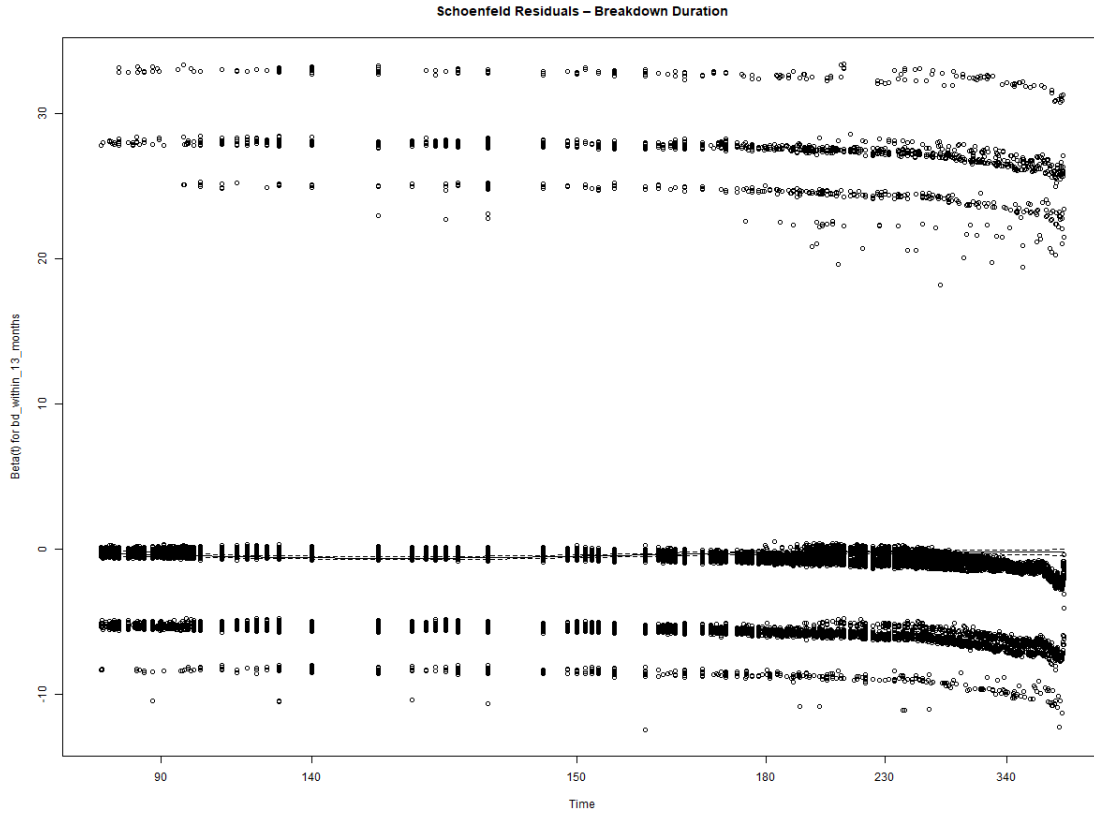


Figure 33: Scaled Schoenfeld residuals for the Cox proportional hazards model of breakdown duration. Systematic trends in residuals over time indicate non-proportional hazards, particularly for cluster and prior breakdown history covariates, reflecting the heterogeneous clearance trajectories of Cluster 3 herds over extended follow-up.

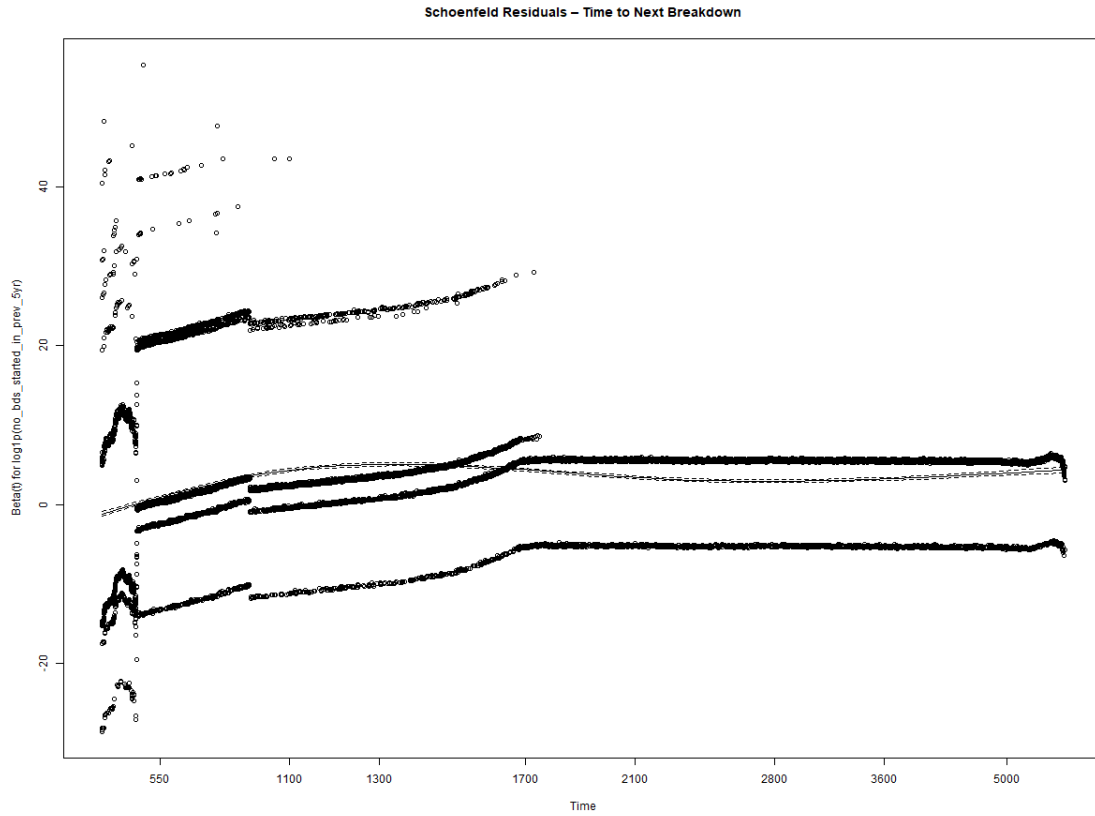


Figure 34: Scaled Schoenfeld residuals for the Cox proportional hazards model of time to subsequent breakdown. Systematic trends indicate non-proportional hazards across covariates, consistent with the heterogeneous recurrence trajectories of the three breakdown phenotypes over extended follow-up.

Supplementary Materials 2

Gaussian Mixture Model

As an additional methodological comparator, Gaussian Mixture Model (GMM) clustering was applied to the standardised feature matrix using the `mclust` package, with the optimal number of components and covariance structure selected according to the Bayesian Information Criterion (BIC). The optimal GMM solution was a two-component model with an ellipsoidal, equal-shape covariance structure (VEV), comprising components of size $n = 13,553$ and $n = 26,120$. The Adjusted Rand Index (ARI) between the GMM two-component hard assignment and the canonical PAM three-cluster solution was 0.784, indicating substantial but not perfect agreement.

The preference of GMM for two components rather than three is consistent with the overlapping and continuous nature of bTB breakdown dynamics, in which the boundary between Cluster 1 (low-intensity) and Cluster 2 (chronic recurrence) phenotypes is less geometrically distinct than the separation of Cluster 3 (acute high-intensity transmission). The GMM result (see Supplementary Material 2 Figure 5.8, Figure 24) does not contradict the three-cluster PAM solution, but rather reflects the different sensitivity of parametric model-based clustering to ellipsoidal cluster boundaries relative to the non-parametric, distance-based approach of PAM. Compared with K-means and PAM, this solution represents a higher-level partitioning of the data, capturing the dominant underlying structure rather than finer subgroups.

As a probabilistic approach, GMM also quantifies uncertainty in cluster membership, highlighting regions of overlap where observations cannot be clearly assigned to a single group. While this provides a more flexible and realistic representation of the data, the two-cluster solution was considered overly coarse for the study objectives. Therefore, the three-cluster solution obtained from the partitioning methods was retained for interpretation.

2. Fuzzy C-means

Fuzzy C-means clustering was applied to the standardised feature matrix with $c = 3$ centres and five random initialisations to examine whether cluster boundaries were discrete or represented a continuum of breakdown behaviour. Unlike PAM, which assigns each observation to exactly one cluster, Fuzzy C-means assigns each observation a vector of partial membership probabilities summing to one, enabling direct quantification of assignment ambiguity.

The ARI between the fuzzy hard assignment, defined as assignment to the cluster with the highest membership probability, and the canonical PAM solution was 0.326, indicating substantial disagreement between the two methods in terms of the specific observations

assigned to each cluster. This divergence reflects a genuine methodological difference rather than instability in the data: Fuzzy C-means optimises a weighted within-cluster dispersion criterion that treats overlapping boundaries as continuous, whereas PAM partitions the space by minimising distances to discrete medoids. The two methods therefore impose different geometric assumptions on the same underlying data.

The distribution of maximum membership probabilities (Supplementary Material 2 Figure 35) revealed substantial heterogeneity in assignment confidence across observations. The median maximum membership probability was 0.671, with the 10th and 90th percentiles at 0.422 and 0.880, respectively, indicating that a considerable proportion of observations had maximum membership probabilities below 0.80, the conventional threshold for confident assignment. The first quartile of 0.513 confirms that at least 25% of observations had near-equal partial membership across two or more clusters, consistent with the overlapping and continuous nature of bTB breakdown dynamics identified in the PCA visualisation.

These findings indicate that the three breakdown phenotypes, while epidemiologically meaningful and structurally stable under PAM, do not correspond to sharply bounded, fully discrete groupings in the feature space. This finding is biologically plausible given the spectrum of breakdown severity and persistence observed in national surveillance data.

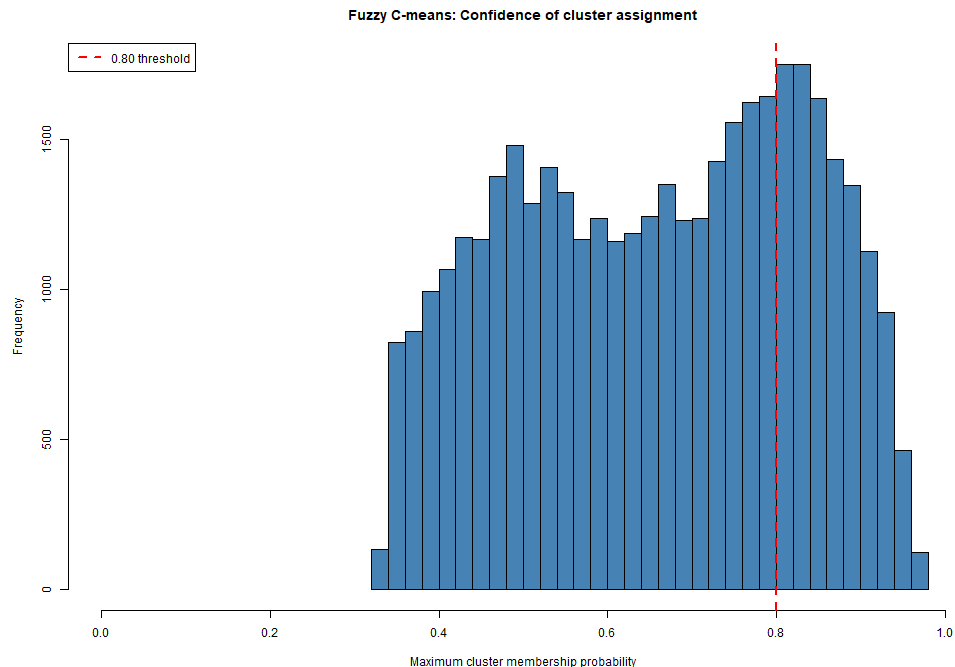


Figure 35: Distribution of maximum cluster membership probabilities from fuzzy C-means clustering ($c = 3$, five random initialisations). Values near 1.0 indicate confident, unambiguous cluster assignment; values near 0.33 indicate near-equal partial membership across all three clusters. The dashed red line marks the 0.80 threshold for confident assignment. The median maximum membership probability was 0.671, indicating that a substantial proportion of breakdown events exhibit partial membership across phenotype boundaries.

Consensus Clustering

Consensus clustering was implemented to provide a pairwise co-occurrence-based assessment of cluster stability, complementary to the Jaccard-based bootstrap stability assessed in Section 3.4.5. Rather than measuring whether the same cluster sets reform after resampling, consensus clustering records, for every pair of observations, the proportion of resampling iterations in which both observations were drawn and assigned to the same cluster, producing a full $N \times N$ consensus matrix whose block-diagonal structure provides visual evidence of genuine cluster boundaries.

The analysis was conducted on a random subsample of $n = 2,000$ observations, consistent with the computational constraints of the $O(n^2)$ matrix construction step, across $B = 200$ resampling iterations, each drawing 80% of observations without replacement. PAM with $K = 3$ was re-applied in each iteration.

The consensus matrix (Figure 36) exhibited a clear block-diagonal structure upon reordering by hierarchical clustering of the consensus distance matrix, with three well-defined blocks corresponding to the three breakdown phenotypes. Off-diagonal consensus values showed a strongly bimodal distribution, with a first quartile of 0.000 and a third quartile of 1.000, indicating that most observation pairs were either consistently co-assigned or consistently separated across resamples, the hallmark of a stable, well-separated cluster solution.

The consensus CDF area-under-the-curve was 0.513, which lies close to the theoretical maximum for a perfectly bimodal distribution, reflecting near-complete polarisation of consensus values toward 0 and 1.

The ARI between the consensus-derived groupings, obtained by cutting the hierarchical clustering of the consensus matrix at $K = 3$, and the canonical PAM solution was 0.941, confirming near-perfect agreement between the two independent clustering approaches and providing strong corroborating evidence for the structural validity of the three-cluster PAM solution.

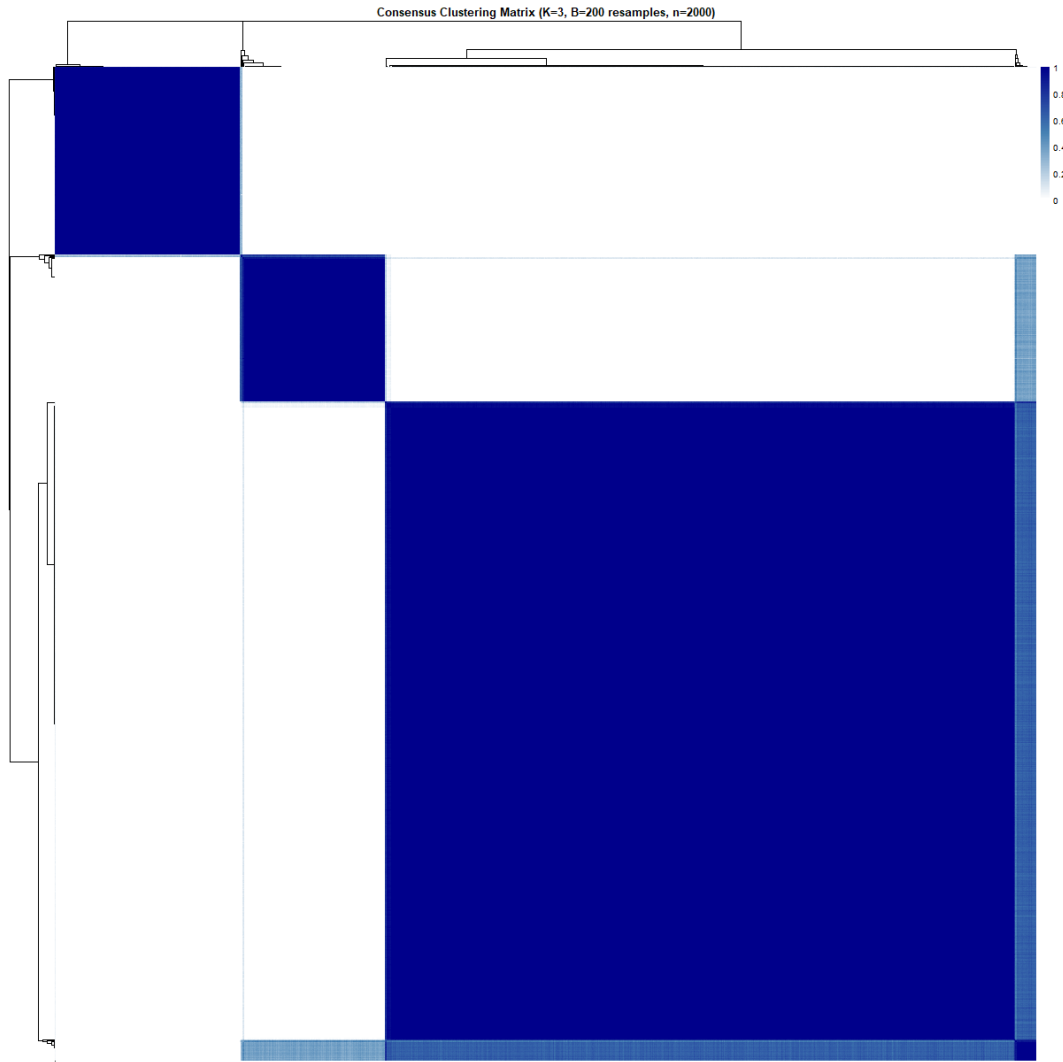


Figure 36: Consensus clustering matrix for PAM $K = 3$, computed across $B = 200$ resampling iterations on a subsample of $n = 2,000$ observations. Each cell represents the proportion of iterations in which the corresponding pair of observations was co-assigned to the same cluster. Rows and columns are reordered by hierarchical clustering of the consensus distance matrix. Clear block-diagonal structure indicates stable, well-separated cluster boundaries.

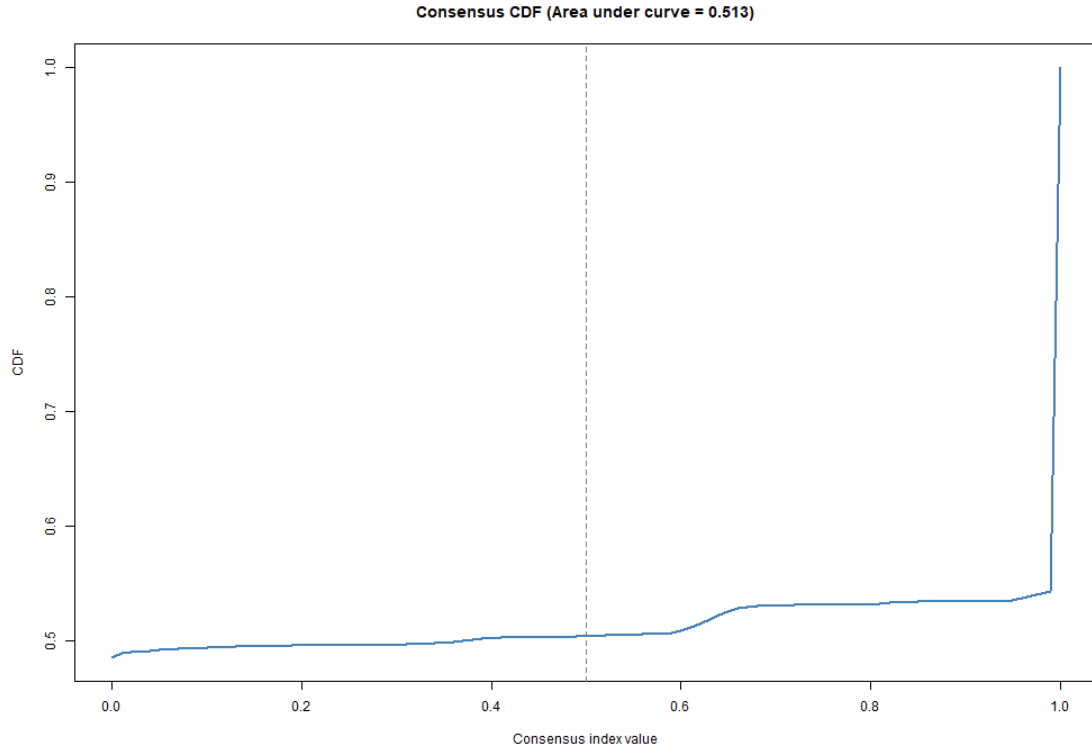


Figure 37: Empirical cumulative distribution function (CDF) of off-diagonal consensus matrix values. The area under the CDF curve was 0.513, reflecting a strongly bimodal distribution of consensus values concentrated near 0 and 1, consistent with stable and well-separated cluster structure.